

TOBB ETÜ

Ekonomi ve Teknoloji Üniversitesi

MAK473 Unmanned Aircraft Design

# Final Report

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# Contents


**1 Introduction** **3**
- 1.1 Executive Summary . . . 3
- 1.2 Requirements . . . 3
- 1.3 Competitor Study . . . 4

**2 Aircraft Layout** **5**
- 2.1 Engine Configuration . . . 6
- 2.2 Wing Layout . . . 8
- 2.3 Empennage Layout . . . 9
- 2.4 Fuselage Layout . . . 9
- 2.5 Landing Gear Configuration . . . 9

**3 Initial Weight Estimation** **10**

**4 Constraint Analysis** **13**
- 4.1 Step 1: Raw Constraints . . . 13
- 4.2 Step 2: Altitude Normalisation . . . 14
- 4.3 Step 3: Takeoff Weight Normalisation . . . 14

**5 Wing Sizing** **15**
- 5.1 Reference Area and Planform . . . 15

**6 Airfoil Selection and High-Lift Device Sizing** **16**
- 6.1 Step 1: Thickness Ratio . . . 16
- 6.2 Step 2: Design Lift Coefficient . . . 16
- 6.3 Step 3: Required Maximum Lift Coefficient . . . 16
- 6.4 Step 4: Candidates and XFLR5 Analysis . . . 17
- 6.5 Step 5: Polar Discussion . . . 17
- 6.6 Step 6: Final Selection . . . 20
- 6.7 Step 7: High-Lift Device . . . 20

**7 Fuselage and Empennage Sizing** **21**
- 7.1 Fuselage Length . . . 21
- 7.2 Diameter . . . 21
- 7.3 Tail Volume Coefficient Method . . . 21
- 7.4 Tail Arm Optimisation . . . 22
- 7.5 Tail Areas and Geometry . . . 22
- 7.6 Tail Airfoil . . . 22

**8 Propeller Sizing** **23**
- 8.1 Number of Blades . . . 23
- 8.2 Diameter . . . 23
- 8.3 Tip Mach Number and Ground Clearance . . . 23

**9 Primary Control Surface Sizing** **24**
- 9.1 Ailerons . . . 24
- 9.2 Elevator . . . 24
- 9.3 Rudder . . . 24

**10 Design Iteration** **24**
10.1 Component Weight Build-Up . . . . . 24
10.2 Gross Weight Iteration . . . . . 26
10.3 Converged Geometry . . . . . 28
10.4 OpenVSP Three-Dimensional Model . . . . . 28
10.5 Parasite Drag from OpenVSP . . . . . 31
10.6 Flow5 Model and Analysis Settings . . . . . 33
10.7 Drag Matching and Maximum Aerodynamic Efficiency . . . . . 34
10.8 Aerodynamic Iteration and Fuel Requirement . . . . . 36

**11 Aircraft Assembly, Trim and CG** **37**
11.1 Wing Incidence . . . . . 37
11.2 Horizontal-Tail Incidence and Cruise Trim . . . . . 37
11.3 Mass Distribution and CG . . . . . 38
11.4 CG Envelope and Static Margin . . . . . 39

**12 Requirement Compliance Analysis** **40**
12.1 Stall Speeds . . . . . 40
12.2 Climb and Ceilings . . . . . 41
12.3 Endurance and Range . . . . . 41
12.4 Takeoff and Landing . . . . . 41
12.5 Flight Envelope and Gust-Load Analysis . . . . . 42
12.6 Compliance Matrix . . . . . 47
12.7 Triple Redundancy Summary . . . . . 47

**13 Conclusion** **50**

# 1 Introduction

## 1.1 Executive Summary

This report documents the complete conceptual design process for a Medium-Altitude Long-Endurance (MALE) class Unmanned Aerial Vehicle sized against an eleven-point Request for Proposal. Design and sizing iterations follow Raymer's *Aircraft Design: A Conceptual Approach* [1]. Throughout the project, critical engineering decisions and reference data were subjected to a "Triple Redundancy" principle — cross-validation from at least three independent sources — to expose the errors and uncertainties that a single source would have hidden.

The aircraft was first sized from historical and statistical relations (Sections 3–9), then reconverged in Section 10 using a component-by-component weight build-up, an OpenVSP full-aircraft parasite-drag analysis, and a three-dimensional Flow5 lifting-surface analysis. The final converged baseline is:

| | | | |
|---|---|---|---|
| Gross takeoff mass $m_0$ | **1464.5 kg** | Reference wing area $S_{ref}$ | **27.683 m$^2$** |
| Operating empty weight (OEW) | 1002.7 kg | Wingspan $b$ | 23.53 m |
| Usable fuel + trapped (5%) | 311.8 kg | Aspect ratio $AR$ | 20 |
| Payload | 150.0 kg | Installed power | 220 HP, tractor |
| $C_{D0}$ (cruise, OpenVSP) | 0.01605 | $(L/D)_{max}$ (cruise, Flow5) | 31.42 |

Two results define the project. First, the statistical empty-weight estimate was confirmed by the component build-up to within 0.4%, and the gross weight converged from 1483.06 kg to 1464.50 kg — a difference of only 1.25%. Second, the assumed parasite drag coefficient of 0.025 proved pessimistic by 36%; the real geometry is substantially cleaner, which raises $(L/D)_{max}$ from 22.42 to 31.42 and releases roughly 65 kg of fuel capacity as margin against the 24-hour endurance requirement.

A third outcome was not anticipated. During three-dimensional assembly the originally selected pusher propeller was found to interfere physically and aerodynamically with the T-tail, forcing a revision to a tractor installation with a chin-mounted payload. That change turned out to fix a latent centre-of-gravity problem as well, and is documented in Sections 2.1 and 11.

## 1.2 Requirements

The table below lists the requirements given by the RFP. Every sizing decision in this report must satisfy these numbers.

| Requirement / Constraint | Target Value | Unit | Engineering Note |
|---|---|---|---|
| Payload | $\geq 150$ | kg | Fundamental parameter for fuselage volume and $m_0$ sensitivity. |
| Landing stall speed | $\leq 40$ | KCAS | Sets the wing area through $C_{L,max}$ in landing configuration. |
| Clean stall speed | $\leq 50$ | KCAS | Clean configuration. |
| Service ceiling | $> 24000$ | ft | Minimum-climb constraint; mandates turbocharging. |
| Cruise ceiling | $> 18000$ | ft | Operational mission altitude. |
| Cruise speed | 70 | KCAS | Reference speed for endurance and range analyses. |
| Endurance | $> 24$ | h | At cruise altitude; the primary constraint driving fuel fraction. |
| Operation radius | $> 2000$ | km | One-way range excluding climb credit. |
| Total mission cruise range | 4000 | km | – |
| Max ROC at sea level | $> 1200$ | fpm | At sea level; determines installed power. |
| Takeoff distance | $< 2000$ | m | Optional constraint, under ISA +30°C conditions. |

Table 1: RFP requirements summary

Three of these eleven requirements do all of the sizing work: the **landing stall speed** fixes the wing loading and therefore the wing area; the **sea-level rate of climb** fixes the installed power; and the **24-hour endurance** fixes the fuel fraction. The remaining eight are satisfied as consequences, and the compliance analysis of Section 12 confirms this.

## 1.3 Competitor Study

The RFP names the platform class but specifies no configuration; the design is therefore driven by the eleven numerical requirements alone, and no obligation exists to resemble any existing aircraft. Competitor data is used here for one purpose only: to bound the initial design space and to sanity-check intermediate results, never as a target to copy. With that caveat, five operational MALE-class UAVs (Bayraktar TB2, TAI Anka-S, IAI Heron, MQ-1 Predator, MQ-9 Reaper) were analysed.

### a) General features

| UAV Model | Origin | Configuration | Propulsion | Mission Profile |
|---|---|---|---|---|
| Bayraktar TB2 | Turkey | Inverted V-tail, pusher | Piston (100 hp) | Tactical/MALE (ISR/strike) |
| TAI Anka-S | Turkey | V-tail, pusher | Piston (170 hp) | MALE (ISR/strike) |
| IAI Heron | Israel | Twin-boom, pusher | Piston (115 hp) | MALE (ISR) |
| MQ-1 Predator | USA | Inverted V-tail, pusher | Piston (115 hp) | MALE (ISR/strike) |
| MQ-9 Reaper | USA | Y-tail, pusher | Turboprop (900 shp) | HALE/heavy MALE (strike) |

### b) Geometrical parameters

| UAV Model | Length [m] | Wingspan [m] | Wing area [m$^2$] | AR |
|---|---|---|---|---|
| Bayraktar TB2 | 6.50 | 12.00 | $\sim 7.0$ | $\sim 20.5$ |
| TAI Anka-S | 8.60 | 17.50 | 13.60 | 22.5 |
| IAI Heron | 8.50 | 16.60 | 13.00 | 21.2 |
| MQ-1 Predator | 8.22 | 14.80 | 11.50 | 19.0 |
| MQ-9 Reaper | 11.00 | 20.00 | 23.50 | 17.0 |

### c) Weight breakdown

| UAV Model | MTOW [kg] | Empty [kg] | Payload [kg] | Fuel [kg] |
|---|---|---|---|---|
| Bayraktar TB2 | 700 | 420 | 150 | 300 |
| TAI Anka-S | 1700 | $\sim 1100$ | 350 | $\sim 250$ |
| IAI Heron | 1150 | 680 | 250 | $\sim 220$ |
| MQ-1 Predator | 1020 | 512 | 204 | $\sim 300$ |
| MQ-9 Reaper | 4760 | 2223 | 1701 | 1800 |

**d) Performance**

| UAV Model | Cruise [KCAS] | Stall [KCAS] | Service ceiling [ft] | Endurance [h] |
|---|---|---|---|---|
| Bayraktar TB2 | 70 | 40 | 25,000 | 27 |
| TAI Anka-S | 110 | 45 | 30,000 | 30 |
| IAI Heron | 80 | $\sim 45$ | 35,000 | 45+ |
| MQ-1 Predator | 70 | 54 | 25,000 | 24 |
| MQ-9 Reaper | 170 | $\sim 75$ | 50,000 | 27 |

**e) Requirements comparison matrix**

| Requirement | RFP Target | Best Competitor Match | Critical Engineering Deduction |
|---|---|---|---|
| Payload | $\geq 150$ kg | 350 kg (Anka-S) | Aircraft sized like MQ-1 or TB2 comfortably meet 150 kg. |
| Endurance | $> 24$ h | 45 h (Heron) | Fuselage volume and $L/D$ are the primary drivers. |
| Stall speed | $\leq 40$ KCAS | 40 KCAS (TB2) | Requires large wing area and an effective flap system. |

**Critical data analysis.** The principal problem with open-source UAV data is that published performance figures are often marketing-oriented [6]. For the MQ-1 Predator the sum $m_e + m_p + m_f = 512 + 204 + 300 = 1016$ kg sits just under the declared MTOW of 1020 kg, which is self-consistent. For the Bayraktar TB2, however, 420+150+300 = 870 kg exceeds the declared MTOW of 700 kg by a wide margin. The physical explanation is that maximum payload and maximum fuel cannot be carried on the same flight the payload–range trade-off. Regressing blindly on such numbers would have corrupted the design space before any calculation began. Each aircraft was therefore verified from three independent sources: manufacturer technical documentation, international defence encyclopaedias such as Jane's [6], and independent academic weight analyses in Raymer [1] and Sadraey [2]; the tables above report reconciled operational values.

# 2 Aircraft Layout

The configuration was selected with a morphological matrix followed by Pugh trade studies. The table below is the *final* configuration, including the propulsion revision made during three-dimensional integration (Section 2.1).

| Configuration Parameter | Selection | Primary Reason |
|---|---|---|
| Propulsion type | Piston (turbocharged) | 24 h endurance demands low SFC; turboprop SFC too high. |
| Number of engines | Single | Minimum $C_{D0}$ penalty; redundancy moved to subsystems. |
| Engine position | **Tractor** (nose) | Revised in Study 05: aft propeller fouls the T-tail. |
| Payload installation | **Chin fairing** | Consequence of the tractor revision; MQ-9 precedent. |
| Wing vertical location | **Shoulder / high wing** | Pendulum stability, debris clearance, clean lower hemisphere. |
| Wing sweep / taper | $\Lambda_{LE} = 0°$, $\lambda = 0.45$ | $M_{cr} = 0.15$; sweep gives no benefit, only weight. |
| Tail type | **T-tail** | Keeps HT clear of wing downwash and fuselage boundary layer. |
| Fuselage layout | **Single-boom** (podded) | Lower wetted area than twin-boom at this fineness ratio. |
| Landing gear | Tricycle, retractable | Runway steering; retraction for cruise $C_{D0}$. |
| Material | Composite primary structure | Empty-weight reduction, Section 10.1. |
| High-lift device | Single-slotted flap | Meets $\Delta C_{l,max}$ with minimum mechanism weight. |

Table 2: Final configuration summary

## 2.1 Engine Configuration

**Piston versus turboprop.** A 24-hour endurance mission is dominated by specific fuel consumption. A modern turbocharged piston engine operates at $c_{bhp} \approx 0.4$–$0.5$ lb/hp/hr, whereas a small turboprop in this power class is typically 0.6–0.7 lb/hp/hr. At constant fuel weight that is directly a 30–40% endurance penalty, so the piston engine was selected. All five competitor aircraft below 2000 kg MTOW use the same solution — an independent third confirmation.



Figure 1: Continental TSIO-360 Series Engine

**Number of engines.** A Pugh matrix trade study was conducted against a single-engine

datum. Quad-motor layouts were excluded immediately because of the parasite-drag penalty of four nacelles on a 28 $m^2$ wing.

| **Criteria** | **Single (1×190 HP)** | **Twin (2×95 HP)** | **Tri (3×65 HP)** |
|---|---|---|---|
| Parasite drag ($C_{D0}$ penalty) | S (datum) | −1 | −2 |
| System dry weight | S (datum) | −1 | −2 |
| Propeller / propulsive efficiency | S (datum) | −1 | −1 |
| Redundancy match | S (datum) | +1 (dual) | +2 (triple) |
| Survivability / fail-safe | S (datum) | +1 | +2 |
| **Net score** | **0 (SELECTED)** | −1 | −1 |

Table 3: Propulsion architecture Pugh matrix

Multi-engine layouts improve survivability, but mounting additional nacelles on a wing of this area compromises the parasite-drag budget and therefore the endurance requirement directly. A single-engine architecture is selected, and the redundancy requirement is met at subsystem level instead (see below).

**Tractor versus pusher — a design revision.** The initial conceptual layout selected a *pusher* installation, on the reasoning that an aft propeller leaves the nose completely unobstructed for an ISR payload, and that all five competitor aircraft are pushers. During Study 05 the complete single-boom fuselage, T-tail and 1.90 m propeller disk were assembled in OpenVSP, and the aft region was found to be geometrically congested: with a tail arm of 4.58 m the propeller disk and the vertical-tail root compete for the same fuselage stations, and the T-tail surfaces would operate inside the propeller slipstream — exactly the condition the T-tail was chosen to avoid.

The propulsion architecture was therefore revised to a **tractor** installation. The engine class, 220 HP installed rating, three-blade propeller and 1.90 m diameter are all retained; only the station changes. To preserve an unobstructed field of regard, the 150 kg EO/IR payload is relocated to a chin fairing immediately below and aft of the engine cowl, which is the standard solution on the MQ-9 Reaper.

This is recorded as a design iteration rather than concealed, because it is an outcome of three-dimensional integration that could not have been seen in the two-dimensional sizing phase. It also had an unforeseen benefit: moving 238 kg of engine group from the tail to the nose removed the centre-of-gravity problem that the pusher layout would have created (Section 11.4).

**Selected engine: Continental TSIO-360.** The constraint analysis of Section 4 requires 190.5 HP of sea-level-equivalent shaft power. With a 15% installation and growth margin the installed rating is set at **220 HP**. To eliminate the severe power lapse at the 24,000 ft service ceiling ($\alpha = 0.393$), a turbocharged and intercooled engine is mandatory.

- Engine: Continental TSIO-360 series (turbocharged and intercooled)
- Required shaft power 190.5 HP; installed rating 220 HP (+15% margin)
- Bare-engine mass 145 kg (320 lb); installed mass 238.3 kg via Raymer's installation factor $m_{eng,inst} = 2.575\, m_{eng}^{0.922}$
- Alternator 12 V, 70 A; TCM S6LSC-25P pressurised magneto with impulse coupling

- Factory-integrated turbocharger; 9.82-inch-core oil cooler; shielded TCM harness with 18 mm spark plugs

**Integration of triple redundancy.** Because a single engine was selected to preserve aerodynamic efficiency, the redundancy requirement is implemented in the subsystems rather than by duplicating nacelles:

1. **Triple-channel FADEC/ECU** powered by three independent electrical buses (main alternator, primary backup battery, emergency avionics battery).
2. **Redundant fuel and ignition paths** using triple high-pressure diaphragms and independent pumps, eliminating single-point failures.
3. **Control-surface segmentation:** elevators and ailerons are split into three independently actuated segments, so a single actuator failure leaves controllable authority.

## 2.2 Wing Layout

**Aspect ratio.** $AR = 20$ was *pre-selected* as a design input, not derived. The reasoning is that induced drag varies as $1/AR$ and the endurance mission is dominated by $L/D$; the five competitor aircraft cluster between $AR = 17$ and 22.5, which bounds the choice. This value was then carried into the constraint analysis as a fixed input. The structural cost of the choice is quantified in Section 10.1, where the wing emerges as the heaviest single structural item. A formal $AR$ trade study was not performed and is recorded as a limitation in Section 13.

**Vertical position.** A shoulder/high-wing layout was selected for three independent reasons: it keeps the wing clear of debris on semi-prepared strips and gives good ground clearance; it produces a favourable pendulum (dihedral-equivalent) lateral stability effect, reducing the geometric dihedral needed on an already very high aspect ratio wing; and it leaves the lower fuselage hemisphere free for the payload turret. In the final CAD model the wing chord plane sits at $z = +0.50$ m against a fuselage radius of 0.66 m, i.e. at 76% of the radius — a shoulder wing rather than a wing mounted on top of the fuselage.

**Sweep and taper.** Two parameters remain free once $AR$ is fixed:

- $\Lambda_{LE} = 0°$: the aircraft cruises at 70 KCAS ($M = 0.15$), far below the speed at which sweep reduces compressibility drag. Sweep would add structural weight and aeroelastic complexity for no benefit.
- $\lambda = 0.45$: taper ratio controls how the chord shrinks from root to tip, and a well-chosen value brings the spanwise loading close to elliptical, minimising induced drag for a given area.

Triple redundancy check for $\lambda$: (1) Raymer [1], Table 4.1 and Fig. 4.19, gives $\lambda \approx 0.3$–$0.5$ for an unswept high-$AR$ wing; (2) Roskam [4], Part II, an independent textbook with its own statistical tables, gives a similar range for the same class; (3) classical aerodynamic theory [3] shows that $\lambda \approx 0.40$–$0.45$ on an unswept wing produces a loading very close to elliptical.

Note that with $\Lambda_{LE} = 0°$ and $\lambda = 0.45$ the quarter-chord sweep is not exactly zero but $\Lambda_{c/4} = -1.09°$ (slightly forward). Since $\cos(1.09°) = 0.9998$, every $\cos \Lambda_{c/4}$ correction in

Section 6 is unaffected and is taken as unity.

## 2.3 Empennage Layout

A T-tail is selected, with the horizontal tail at the fin tip. This keeps the horizontal tail clear of the strongest part of the wing wake and out of the fuselage boundary layer, so it retains authority throughout the envelope. In the original pusher layout the T-tail also served to keep the horizontal surface out of the propeller slipstream; after the revision to a tractor installation that particular argument no longer applies, and the retained justification is wake and boundary-layer avoidance plus a clean aft structural join.

The penalties are a heavier vertical tail — the fin must carry the horizontal surface, which is why the weight relation is applied to the full fin area — and a deep-stall risk. The latter is mitigated by the gentle stall characteristics of the selected NACA 4415 section ($\alpha_{stall} \approx 17^\circ$ with gradual post-stall lift decay) and by an angle-of-attack limiter in the flight control law.

## 2.4 Fuselage Layout

A single-boom (podded) fuselage was selected over a twin-boom layout. At this fineness ratio a single body has significantly lower wetted area than two booms plus a central pod, and $C_{D0}$ scales with wetted area, which matters directly for endurance. The fuselage diameter follows from the fineness ratio $f = L/d$; Raymer [1] recommends $f \approx 4$–$8$ for low drag and $f = 6$ is selected. A circular cross-section is assumed, so width and height both equal the diameter.

## 2.5 Landing Gear Configuration

A retractable tricycle landing gear is selected. Tricycle geometry gives positive nose-wheel steering on the ground, keeps the fuselage level in the ground attitude for sensor operation, and avoids the propeller-strike risk of a taildragger during rotation — important with the 1.90 m tractor propeller of Section 8. Retraction is required to remove the substantial drag contribution of the exposed wheels and struts from the cruise polar; at the endurance $L/D$ this drag would penalise range and endurance directly.

To maintain structural simplicity and protect sensitive optical payload sensors typically mounted under the forward fuselage, a **Tricycle Landing Gear** arrangement with a fixed or semi-retractable main gear setup is selected. The nose gear is positioned at $x = 1.25$ m from the nose, providing an adequate ground clearance ($\approx 1.20$ m at the propeller hub) to prevent propeller tip ground strikes during rotation and landing. The Main Landing Gear (MLG) is located at $x = 2.30$ m, approximately $15^\circ$ to $20^\circ$ aft of the MTOW CG ($x_{CG} = 2.091$ m), strictly satisfying the static tip-over and ground stability criteria.

**Stowage.** The nose gear retracts forward into a bay ahead of the payload chin fairing; the main units retract inboard into the wing root and centre-fuselage keel, taking advantage of the shoulder-wing carry-through structure.

**Longitudinal positioning.** The main gear is placed at $x_{MLG} = 2.30$ m, 0.209 m ($\approx 17\%\bar{c}$) aft of the MTOW CG station of $x_{cg} = 2.091$ m (Section 11.3). This corresponds to a static tip-back angle of approximately $16^\circ$ against the propeller-hub height of 1.18 m,

comfortably above the customary 15° minimum and safely below the maximum useable rotation attitude. The nose gear sits at $x_{NLG} = 1.25$ m, giving a wheelbase of

$$L_{WB} = x_{MLG} - x_{NLG} = 2.30 - 1.25 = 1.05 \text{ m} \tag{1}$$

and a nose-gear load fraction at MTOW of $(x_{MLG} - x_{cg})/L_{WB} \approx 19.9\%$, inside the recommended 8–20% range for a tricycle configuration.

**Track and lateral stability.** The main-gear track is set at $b_{track} = 2.00$ m to give an overturn angle below 55° against the MTOW CG height (fuselage axis at 1.18 m above the ground plus $0.4\,d = 0.53$ m of internal CG offset). This is comfortable for semi-prepared operation.

**Propeller ground clearance.** The dominant sizing constraint is not stability but propeller-tip clearance: the 1.90 m tractor propeller extends 0.95 m below the fuselage axis. Maintaining the customary 0.23 m (9 in) static tip clearance therefore requires the fuselage axis at least 1.18 m above the ground (see Section 7.3), which drives the nose- and main-gear leg lengths. This is the operational cost of the tractor revision of Section 2.1.

**Mass allocation.** The total gear group of 62.97 kg (Section 10.1) is split 30% nose / 70% main — 18.90 kg and 44.07 kg respectively — following the standard tricycle load-carrying split.

# 3 Initial Weight Estimation

**Design assumptions.** To start the takeoff-gross-weight loop, a set of assumptions on flight mechanics, propulsion and weight fractions was required. These were taken from historical MALE-class data and the Raymer methodology and cross-validated against Sadraey [2] and Roskam [4].

| Category | Parameter | Symbol | Value / basis |
|---|---|---|---|
| Aerodynamic | Wing aspect ratio | $AR$ | 20 (pre-selected input, Section 2.2). |
| Aerodynamic | Oswald efficiency | $e$ | 0.8 (conservative for straight, high-$AR$ wings). |
| Aerodynamic | Induced drag factor | $K$ | $1/(\pi e AR) = 0.01989$. |
| Aerodynamic | Zero-lift drag coefficient | $C_{D0}$ | 0.025 (*assumption*; re-computed in Section 10.5). |
| Aerodynamic | Max lift-to-drag ratio | $(L/D)_{max}$ | $1/(2\sqrt{KC_{D0}}) = 22.42$ (*assumption*; Section 10.7). |
| Aerodynamic | Endurance lift-to-drag ratio | $(L/D)_E$ | $0.866\,(L/D)_{max} = 19.42$ (propeller endurance condition). |
| Aerodynamic | Max lift, clean | $C_{L,max,clean}$ | 1.4 (initial threshold; refined to 1.4254 in Section 6). |
| Aerodynamic | Max lift, landing | $C_{L,max,landing}$ | 2.0 (high-lift system deployed). |
| Propulsion | Propeller efficiency, cruise | $\eta_{p,cruise}$ | 0.80 (variable-pitch propeller). |
| Propulsion | Propeller efficiency, climb | $\eta_{p,climb}$ | 0.75. |
| Propulsion | SFC, cruise | $c_{bhp,cruise}$ | 0.4 lb/hp/hr. |
| Propulsion | SFC, loiter | $c_{bhp,loiter}$ | 0.5 lb/hp/hr (part throttle). |
| Mission | Warm-up, taxi, takeoff | $W_1/W_0$ | 0.975 (historical). |
| Mission | Climb | $W_2/W_1$ | 0.980 (historical). |
| Mission | Descent | $W_5/W_4$ | 0.990 (idle descent). |
| Mission | Landing and taxi | $W_6/W_5$ | 0.995 (historical). |
| Mission | Trapped-fuel factor | $F_{trapped}$ | 1.05 (5% unusable). |
| Weight | Payload | $m_p$ | 150 kg (RFP, fixed). |
| Weight | Crew | $m_c$ | 0 kg (unmanned). |

Table 4: Initial design assumptions

*Note on composite structure.* The benefit of composite primary structure is **not** applied at this stage. The historical empty-weight regression of Eq. (3) is used in raw form, and the composite saving is applied later, component by component, in the build-up of Section 10.1. This avoids counting the same saving twice.

**Mission profiles.** Two scenarios covering the operational envelope were built: an *endurance* mission and a *radius* mission.

- Taxi and takeoff: engine start, taxi, takeoff. Historical, 0.975.
- Climb: to 18,000 ft. Historical, 0.980.
- Cruise out: transit to the target area (Breguet range equation).
- Loiter: 24 hours on station (Breguet endurance equation).
- Descent: historical with glide assumption, 0.990.
- Landing: approach and shutdown. Historical, 0.995.

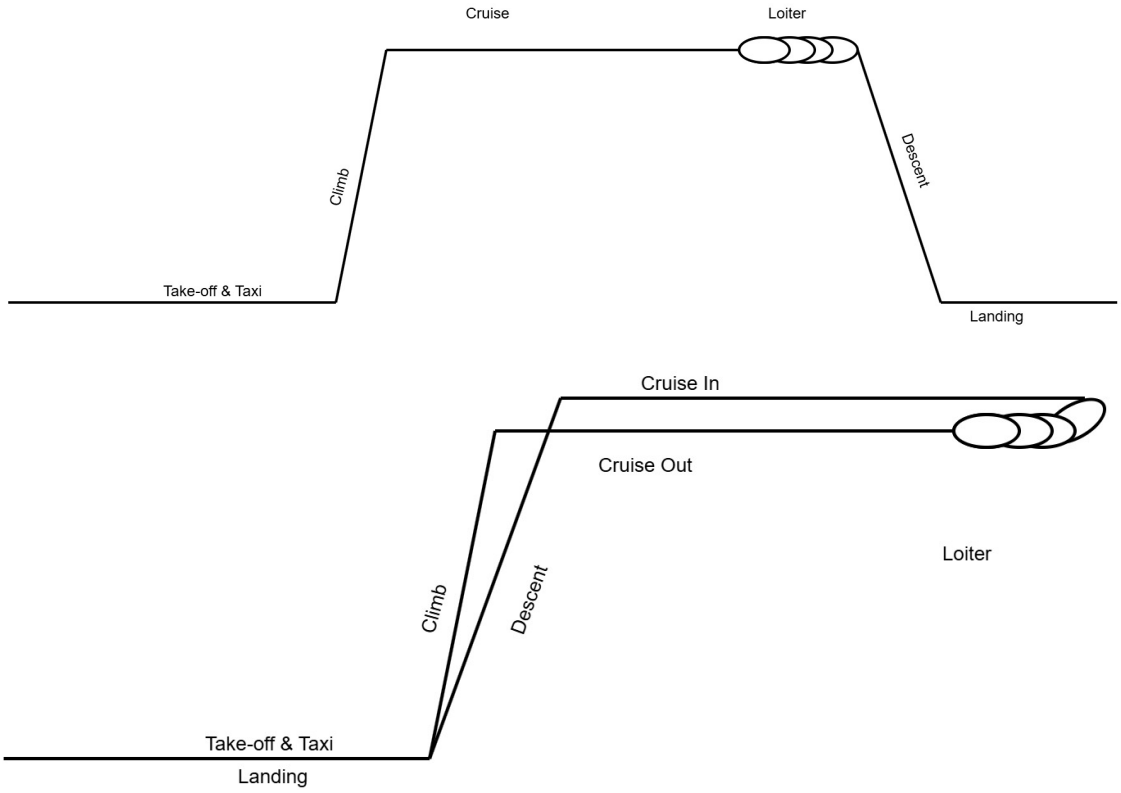



Figure 2: Endurance and Range mission profiles. The 24-hour loiter segment at 18,000 ft is the sizing case.

**Breguet equations (Raymer [1]).** For a propeller-driven aircraft,

$$R = \frac{550\,\eta_p}{c_{bhp}} \left(\frac{L}{D}\right) \ln \frac{W_i}{W_{i+1}} \quad \Longrightarrow \quad \frac{W_{i+1}}{W_i} = \exp\left(\frac{-R\,c_{bhp}}{550\,\eta_p\,(L/D)}\right) \tag{2}$$

$$E = \frac{550\,\eta_p}{V\,c_{bhp}} \left(\frac{L}{D}\right) \ln \frac{W_i}{W_{i+1}} \quad \Longrightarrow \quad \frac{W_{i+1}}{W_i} = \exp\left(\frac{-E\,V\,c_{bhp}}{550\,\eta_p\,(L/D)}\right) \tag{3}$$

The range segment uses $(L/D)_{max}$, the condition for minimum drag; the endurance segment uses $(L/D)_E = 0.866\,(L/D)_{max}$, the condition for minimum *power* required. Confusing the two would cost several hours of predicted endurance.

| **Segment** | **Endurance mission** | **Radius mission** | **Source** |
|---|---|---|---|
| Warm-up / taxi / takeoff | 0.9750 | 0.9750 | Historical |
| Climb | 0.9800 | 0.9800 | Historical |
| Main segment | 0.80263 (24 h loiter) | 0.86259 (4000 km) | Breguet |
| Descent | 0.9900 | 0.9900 | Historical |
| Landing / taxi back | 0.9950 | 0.9950 | Historical |
| Total $W_{final}/W_0$ | 0.75545 | 0.81188 | – |
| $m_f/m_0$ (incl. 5% trapped) | **0.25678** | 0.19752 | – |

Table 5: Mission fuel-fraction comparison. The endurance mission is the sizing case.

**Which mission sizes the aircraft?** With the endurance-sized fuel fraction the achievable cruise range is 5950 km, comfortably above the 4000 km requirement. With the range-sized fuel fraction the achievable endurance would be only 16.1 h, well short of 24 h. The endurance requirement is therefore binding, and the radius requirement is satisfied automatically rather than by over-design.

**Initial gross weight.** A rubber-engine approach was used with payload and mission profile held constant. The empty-weight fraction is modelled with Raymer's historical regression for the UAV/high-altitude class in metric coefficients:

$$\frac{m_e}{m_0} = A\, m_0^C, \qquad A = 2.39,\ C = -0.18 \quad (m_0 \text{ in kg}) \tag{4}$$

$$m_0 = \frac{m_p}{1 - \dfrac{m_f}{m_0} - \dfrac{m_e}{m_0}} \tag{5}$$

This is non-linear because $m_e/m_0$ is itself a function of $m_0$. Starting from 1200 kg and using Excel Goal Seek:

| **Quantity** | **Symbol** | **Value** |
|---|---|---|
| Takeoff gross mass | $m_0$ | **1483.06 kg** (14,543.86 N) |
| Empty-weight fraction | $m_e/m_0$ | 0.6421 |
| Empty mass | $m_e$ | 952.24 kg |
| Fuel fraction | $m_f/m_0$ | 0.25678 |
| Fuel mass | $m_f$ | 380.82 kg |
| Payload | $m_p$ | 150.00 kg |

Table 6: Converged initial weight estimate

**Payload trade study.** The fuel fraction was held constant and the sizing loop re-run for each payload.

| Component | 100 kg | 125 kg | 150 kg (RFP) | 175 kg | 200 kg |
|---|---|---|---|---|---|
| Payload [kg] | 100.0 | 125.0 | 150.0 | 175.0 | 200.0 |
| Fuel [kg] | 319.3 | 350.7 | 380.8 | 410.0 | 438.3 |
| Empty mass [kg] | 824.0 | 890.0 | 952.2 | 1011.6 | 1068.5 |
| $m_e/m_0$ | 0.6628 | 0.6517 | 0.6421 | 0.6336 | 0.6260 |
| **Total $m_0$ [kg]** | **1243.3** | **1365.6** | **1483.1** | **1596.6** | **1706.8** |

Table 7: Payload sensitivity (Raymer historical model, $m_f/m_0 = 0.25678$ fixed)

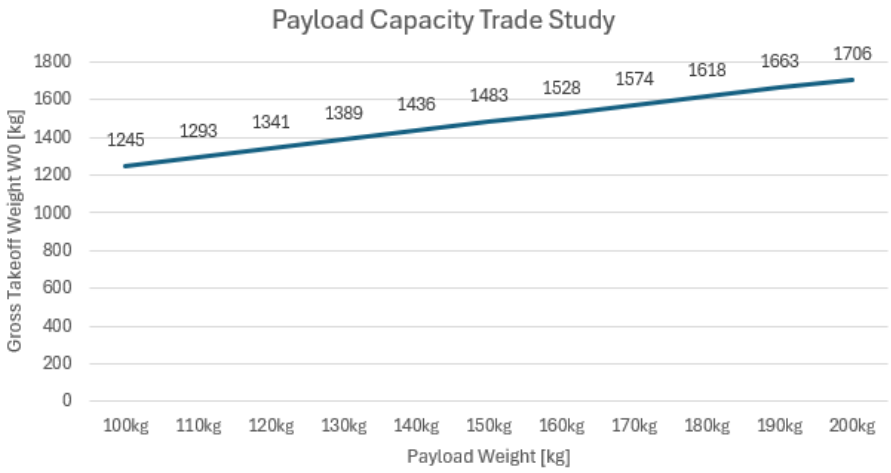



Figure 3: Payload sensitivity. The slope of this curve is the growth factor.

Raising the payload from 150 to 200 kg grows MTOW by 224 kg: a growth factor of about **4.5** kg of aircraft per kg of payload. The factor falls slightly as the aircraft grows because $m_e/m_0$ decreases with $m_0$. The 150 kg design point is adopted — it meets the RFP exactly, without paying the compounding structural penalty of an over-specified payload.

# 4 Constraint Analysis

The objective is to determine the feasible design space and select the wing loading and power loading. The process has three steps: raw constraints, altitude normalisation, and takeoff-weight normalisation. Common constants are $C_{D0} = 0.025$ and $K = 0.01989$, with $AR = 20$ as a fixed input.

## 4.1 Step 1: Raw Constraints

**Stall limit.** The maximum allowable wing loading is set by the 40 KCAS landing stall requirement, and appears as a vertical boundary:

$$\left(\frac{W}{S}\right)_{stall} = \frac{1}{2}\rho_0 V_{stall}^2 C_{L,max} = \tfrac{1}{2}(1.225)(20.578)^2(2.0) = 518.7 \text{ N/m}^2 \tag{6}$$

**Sea-level rate of climb** (1200 fpm = 6.096 m/s):

$$\left(\frac{P}{W}\right)_{SL} = \frac{V_Y}{\eta_p}\left(\frac{q_{SL}C_{D0}}{W/S} + \frac{K}{q_{SL}}\cdot\frac{W}{S}\right) + \frac{ROC_{SL}}{\eta_p} \tag{7}$$

with $V_Y = 27.54$ m/s, $\eta_p = 0.75$, $q_{SL} = \frac{1}{2}\rho_0 V_Y^2$.

**Service ceiling** (24,000 ft, residual 100 fpm = 0.508 m/s):

$$\left(\frac{P}{W}\right)_{24k} = \frac{V_{Y,24k}}{\eta_p}\left(\frac{q_{24k}C_{D0}}{W/S} + \frac{K}{q_{24k}}\cdot\frac{W}{S}\right) + \frac{ROC_{24k}}{\eta_p} \tag{8}$$

with $V_{Y,24k} = 30.52$ m/s.

**Cruise** (18,000 ft, level flight, $ROC = 0$):

$$\left(\frac{P}{W}\right)_{18k} = \frac{V_{cruise}}{\eta_{p,cruise}}\left(\frac{q_{18k}C_{D0}}{W/S} + \frac{K}{q_{18k}}\cdot\frac{W}{S}\right) \tag{9}$$

with $V_{cruise} = 47.71$ m/s TAS and $\eta_{p,cruise} = 0.80$.

## 4.2 Step 2: Altitude Normalisation

Piston engines lose power with altitude. The empirical Gagg & Farrar relation $\alpha = 1.132\sigma - 0.132$ converts each local requirement to a sea-level-equivalent shaft power: $\alpha_{SL} = 1.000$, $\alpha_{18k} = 0.513$, $\alpha_{24k} = 0.393$. Each altitude constraint is divided by its own $\alpha$. The stall limit is unaffected, depending only on wing geometry and sea-level density.

## 4.3 Step 3: Takeoff Weight Normalisation

As fuel burns the weight falls, so all constraints are scaled to takeoff weight with $\beta = W_{local}/W_0$. The inner $\beta$ accounts for the reduced weight actually carried by the wing; the outer $\beta$ returns the ratio to the $W_0$ reference frame. The fractions are $\beta_{SL} = 1.000$ and $\beta_{18k} = \beta_{24k} = 0.9555$.

$$\left(\frac{P}{W_0}\right)_{final} = \frac{\beta}{\alpha}\left[\frac{V}{\eta_p}\left(\frac{q\,C_{D0}}{\beta(W_0/S)} + \frac{K\beta(W_0/S)}{q}\right) + \frac{ROC}{\eta_p}\right] \tag{10}$$

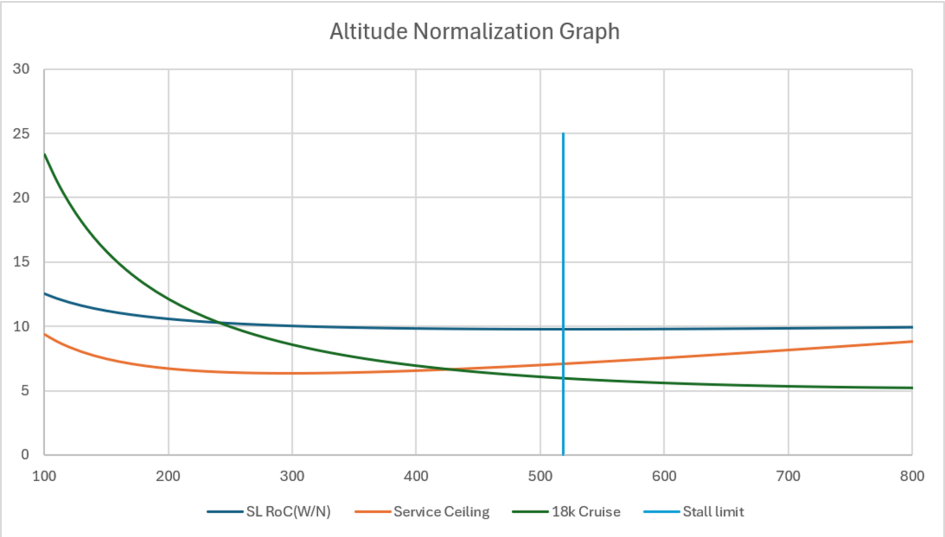



Figure 4: Weight- and altitude-normalised constraint diagram. The design point sits at the intersection of the stall limit and the sea-level climb boundary — the smallest wing and smallest engine that satisfy every requirement simultaneously.

**Design point.** Placing the design point at the intersection of the stall limit and the sea-level ROC boundary minimises both wing area and engine size:

- $(W_0/S)_{opt} = 518.7$ N/m$^2$
- $(P/W_0)_{opt} = 9.77$ W/N

With $W_0 = 14{,}543.86$ N:

$$S_{ref} = \frac{14543.86}{518.7} = 28.04 \text{ m}^2, \qquad P_{req} = 14543.86 \times 9.77 = 142{,}093 \text{ W} = 190.5 \text{ HP} \quad (11)$$

A 15% installation and growth margin gives an installed rating of **220 HP**.

# 5 Wing Sizing

## 5.1 Reference Area and Planform

$$b = \sqrt{AR \cdot S_{ref}}, \qquad c_{root} = \frac{2S_{ref}}{b(1+\lambda)}, \qquad c_{tip} = \lambda\, c_{root} \quad (12)$$

$$\bar{c} = \frac{2}{3} c_{root} \frac{1+\lambda+\lambda^2}{1+\lambda}, \qquad y_{MAC} = \frac{b}{6}\,\frac{1+2\lambda}{1+\lambda} \quad (13)$$

Here $b$ is the span, $c_{root}$ the centreline chord, $c_{tip}$ the tip chord, $\bar{c}$ the mean aerodynamic chord used in all later stability work, and $y_{MAC}$ its spanwise station.

At this stage ($S_{ref} = 28.038$ m$^2$, $\lambda = 0.45$, $AR = 20$):

$$b = \sqrt{20 \times 28.038} = 23.680 \text{ m} \quad (14)$$

$$c_{root} = 1.633 \text{ m}, \qquad c_{tip} = 0.735 \text{ m} \quad (15)$$

$$\bar{c} = 1.241 \text{ m}, \qquad y_{MAC} = 5.172 \text{ m} \quad (16)$$

These values are superseded by the converged geometry of Section 10.4 ($S_{ref} = 27.683$ m$^2$, $b = 23.53$ m, $\bar{c} = 1.233$ m), which is the geometry actually modelled in OpenVSP and Flow5. The difference is 1.3% in area and 0.6% in chord.

Note that the mean *geometric* chord $S/b = 1.177$ m differs from the mean *aerodynamic* chord by about 4.8% because of the taper; the MAC is the correct reference for tail volume coefficients and CG.

# 6 Airfoil Selection and High-Lift Device Sizing

## 6.1 Step 1: Thickness Ratio

Thickness ratio drives wing structural weight: statistical wing-weight relations vary roughly inversely with the square root of $t/c$ [1], so halving $t/c$ increases wing weight by about 41%, which propagates to roughly 6% on empty weight once fed back through the sizing loop.

An $AR = 20$ wing carries large root bending moments, so the section must provide adequate spar depth, internal volume for fuel, and acceptable stall behaviour without an excessive profile-drag penalty. A target of $(t/c)_{max} = 0.15$ was selected. With $\bar{c} = 1.233$ m the maximum physical thickness is $t_{max} = 0.185$ m.

## 6.2 Step 2: Design Lift Coefficient

The aircraft spends the dominant part of its mission in the endurance segment, so the design lift coefficient is evaluated at the average weight of that segment. At the initial sizing weight, $m_i = 1483.06(0.975)(0.98) = 1417.06$ kg, $m_f = 1417.06(0.80263) = 1137.38$ kg, and

$$m_{avg} = 1277.22 \text{ kg}, \qquad W_{avg} = 12{,}525.28 \text{ N} \tag{17}$$

At 18,000 ft ($\rho = 0.69798$ kg/m$^3$, $V_{TAS} = 47.707$ m/s, $S = 28.038$ m$^2$):

$$C_{L_C} = \frac{2W_{avg}}{\rho V_{TAS}^2 S} = 0.5624 \tag{18}$$

Assuming the wing produces 95% of total lift,

$$C_{L_{C_w}} = 0.5920, \qquad C_{l_d} = \frac{C_{L_{C_w}}}{0.9 \cos \Lambda_{c/4}} = 0.6578 \tag{19}$$

Airfoils efficient near $C_l = 0.66$ were therefore preferred. Re-evaluating at the converged mass of 1464.5 kg gives $C_L = 0.5555$ (see Section 11.2), so this target is unchanged in practical terms.

## 6.3 Step 3: Required Maximum Lift Coefficient

From the 50 KCAS clean stall requirement, with $\rho_0 = 1.225$ kg/m$^3$ because KCAS is referenced to sea level:

$$(C_{Lmax})_{base} = \frac{2W_0}{\rho_0 V_{s,clean}^2 S} = 1.2801, \qquad (C_{Lmax})_{wing} = 1.3475 \tag{20}$$

$$C_{lmax,req} = \frac{1.3475}{0.9\cos 0^\circ} = 1.4972 \tag{21}$$

Candidates with $C_{lmax} \geq 1.4972$ were preferred. For reference, the parabolic polar $C_D = C_{D0} + KC_L^2$ peaks at

$$C_{L,opt} = \sqrt{C_{D0}/K} = 1.121, \qquad (L/D)_{max} = \frac{1}{2\sqrt{KC_{D0}}} = 22.42 \tag{22}$$

using the assumed $C_{D0} = 0.025$; this is revised in Section 10.7.

## 6.4 Step 4: Candidates and XFLR5 Analysis

Eight airfoils from six families were screened: NACA 2415, NACA 4415, NACA 23015, NACA $65_2$-415, NACA $63_2$-615, S1223 (12.1%), E423 (12.5%) and FX 63-137 (13.7%, smoothed). All two-dimensional analyses used identical settings: $Re = 2.5 \times 10^6$, $M = 0.15$, $N_{crit} = 9$, $\alpha$ from $-5^\circ$ to $20^\circ$ in $0.5^\circ$ steps, free transition, 200-iteration limit. At 18,000 ft the standard temperature is 252.49 K, giving $a = 318.54$ m/s and $M = 0.150$; with local density, MAC and viscosity the Reynolds number is $2.55 \times 10^6$, rounded to $2.5 \times 10^6$ for a common polar definition.

## 6.5 Step 5: Polar Discussion

**Lift curve.** The high-lift sections produce the largest lift coefficients: E423 reaches $C_{lmax} = 2.088$ near $12^\circ$, FX 63-137 reaches 2.089 near $16^\circ$, and S1223 reaches a maximum converged 2.298 at $11^\circ$ — though several S1223 points fail to converge, so its post-stall behaviour is not reliable. Among the 15%-thick conventional candidates, NACA 4415 reaches $C_{lmax} = 1.667$ near $17^\circ$ with a gradual post-stall decay. NACA 2415 stalls at a similar angle with lower peak lift; NACA 23015 has both lower peak lift and a sharper drop. The 6-series sections do not reach stall within the analysed range.

**Drag polar.** NACA $65_2$-415 has the lowest minimum drag ($C_{d,min} = 0.00450$), followed by NACA $63_2$-615 (0.00503), reflecting the laminar drag bucket of the 6-series. At the design lift coefficient, however, NACA $63_2$-615 gives the highest reliable $L/D$, followed by NACA 4415 and FX 63-137. E423 and S1223 generate very high lift, but their camber costs drag and produces strongly negative pitching moments.

**Efficiency.** Peak section $L/D$ values are 205.21 (E423), 189.57 (FX 63-137), 152.80 (S1223) and 150.94 (NACA 4415). At $C_{l_d} = 0.6578$ the reliable values are 119.47 (NACA $63_2$-615), 108.19 (NACA 4415), 107.53 (FX 63-137), 95.78 (NACA $65_2$-415), 91.54 (NACA 23015), 83.51 (NACA 2415) and 82.44 (E423). The S1223 value at the target $C_l$ is discarded because the converged polar is discontinuous there.

**Pitching moment.** $C_{m0}$ sets the tail download needed for trim and therefore trim drag. NACA 23015 is most favourable ($-0.0040$), then NACA 2415 ($-0.0510$). NACA 4415 has $C_{m0} \approx -0.0925$, more negative than those two but far milder than FX 63-137 ($-0.2187$), E423 ($-0.2428$) and S1223 ($-0.286$), which would demand large tail downloads. Moment behaviour near stall should be treated cautiously, since XFoil-based predictions degrade once flow separates.

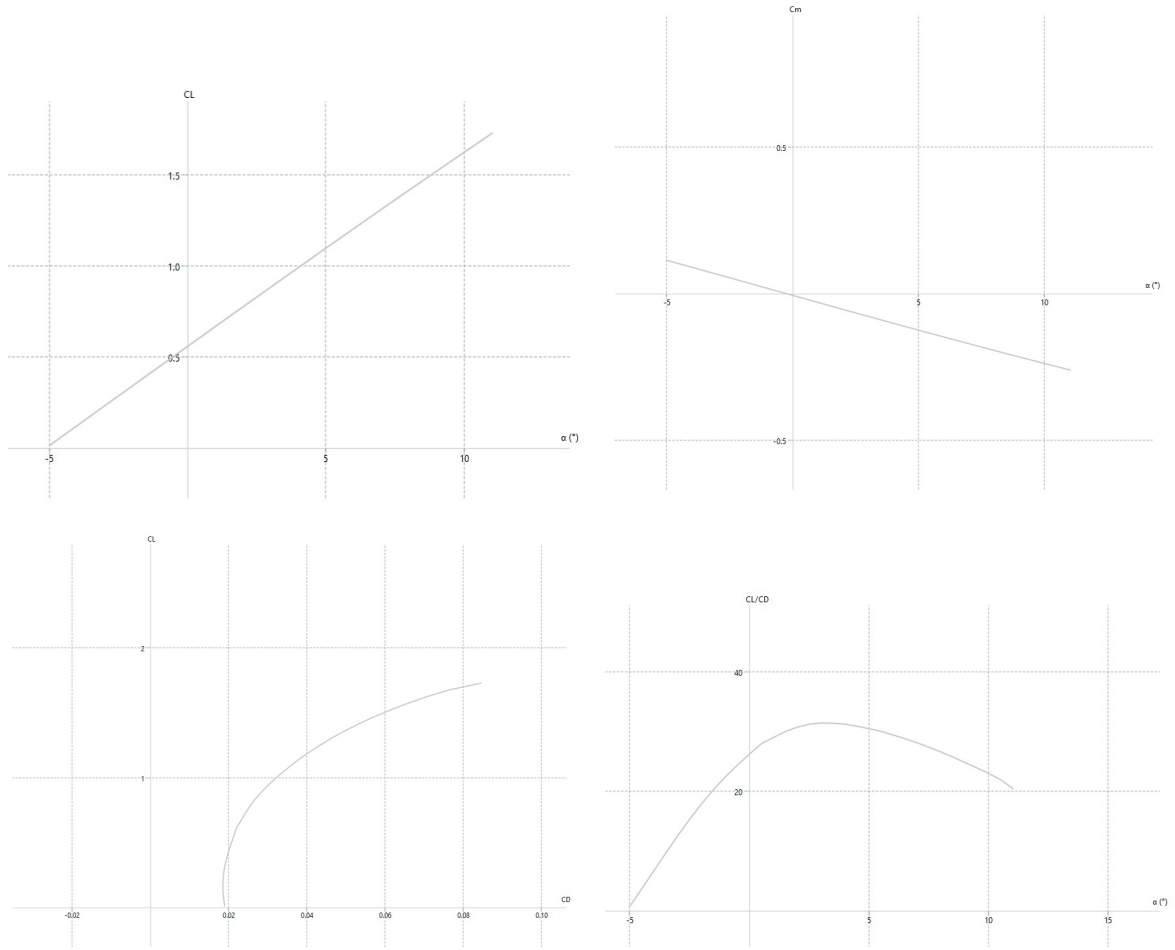



Figure 5: XFLR5 two-dimensional analysis at $Re = 2.5 \times 10^6$, $M = 0.15$: lift curve, pitching moment, drag polar and section efficiency for the candidate airfoils.

| **Airfoil** | **Family** | $C_{l_{max}}$ | $C_{m0}$ | $C_{d_{min}}$ | $(L/D)_{max}$ | $L/D$ at $C_{l_d} = 0.658$ | $\alpha_{stall}$ | $t/c$ | **Stall Quality** |
|---|---|---|---|---|---|---|---|---|---|
| NACA 2415 | NACA 4-digit | 1.640 | −0.0510 | 0.00693 | 96.97 | 83.50 | 17.0° | 0.150 | Gentle-moderate |
| NACA 4415 | NACA 4-digit | 1.667 | −0.0925 | 0.00598 | 150.94 | 108.19 | 17.0° | 0.150 | Gentle |
| NACA 23015 | NACA 5-digit | 1.578 | −0.0040 | 0.00669 | 103.35 | 91.51 | 16.5° | 0.150 | Moderate |
| NACA 65(2)-415 | NACA 6-series | 1.263 | −0.0825 | 0.00450 | 112.76 | 96.11 | $> 20°$ | 0.150 | Not captured |
| NACA 63(2)-615 | NACA 6-series | 1.389 | −0.1193 | 0.00503 | 126.40 | 119.43 | $> 20°$ | 0.150 | Not captured |
| S1223 | Selig | $\geq 2.298^*$ | −0.2860 | 0.00955 | 152.80 | $\approx 48.86^*$ | $> 11°^*$ | 0.121 | Undetermined |
| E423 | Eppler | 2.088 | −0.2428 | 0.00716 | 205.21 | 82.53 | 12.0° | 0.125 | Gentle |
| FX 63-137 smoothed | Wortmann | 2.089 | −0.2187 | 0.00541 | 189.57 | 107.57 | 16.0° | 0.137 | Gentle |

Table 8: Airfoil Performance Comparison Matrix (Analyzed at $Re = 2.5 \times 10^6$, $M = 0.15$)

** Note: S1223 airfoil data is considered unreliable or undetermined at specific points due to XFLR5 operating point convergence failures beyond* 11° *and near the design lift coefficient.*

## 6.6 Step 6: Final Selection

The selected baseline is **NACA 4415**:

- Meets the required maximum lift: $C_{lmax} = 1.6671 > 1.4972$.
- Matches the structural thickness target exactly: $(t/c)_{max} = 15\%$.
- High peak efficiency (150.94) and 108.19 at the actual design $C_l$.
- High stall angle ($\approx 17°$) with gradual post-stall decay — important for a T-tail aircraft, where an abrupt stall could trigger deep stall.
- Pitching moment far milder than the high-lift alternatives, limiting trim drag.

NACA $63_2$-615 is more efficient at the design point but does not reach the required $C_{lmax}$, so it fails the governing screen.

## 6.7 Step 7: High-Lift Device

For a fixed section, $C_{L,max}$ is set by the physics of the flow [3]; reducing $V_{stall}$ further requires artificial high-lift devices.

$$(C_{Lmax})_{wing,clean} = 1.6671 \times 0.9 = 1.5004, \qquad (C_{Lmax})_{clean,A/C} = 1.5004 \times 0.95 = 1.4254 \tag{23}$$

At $V_{s,L} = 40$ KCAS and the initial $W_0$:

$$(C_{Lmax})_{landing,req} = 2.0006, \qquad \Delta C_{Lmax,HLD} = 0.5753 \tag{24}$$

With

$$\Delta C_{Lmax,HLD} = 0.9\,\Delta C_{l,max} \left(\frac{S_{flapped}}{S_{ref}}\right) \cos\Lambda_{HL} \tag{25}$$

and inboard trailing-edge flaps over 60% of the reference area with an unswept hinge line,

$$\Delta C_{l,max,req} = \frac{0.5753}{0.9 \times 0.60} = 1.0653 \tag{26}$$

- Plain / split flaps: $\Delta C_{l,max} \approx 0.90$ — insufficient.
- Single-slotted flap: $\Delta C_{l,max} \approx 1.30$ — satisfies the requirement with margin.
- Fowler / double-slotted: $\geq 1.60$ — capable, but heavy track mechanisms and parasitic drag are unsuitable for an endurance UAV.

**Single-slotted flap selected.** At a takeoff setting ($\delta_f \approx 15$–$20°$), trailing-edge devices deliver about 70% of the landing increment:

$$\Delta C_{Lmax,TO} = 0.4914, \qquad (C_{Lmax})_{TO} = 1.9168 \tag{27}$$

The full landing value is $(C_{Lmax})_L = 1.4254 + 0.702 = 2.1274$, used in all stall and landing calculations.

# 7 Fuselage and Empennage Sizing

## 7.1 Fuselage Length

No first-principles formula exists at this stage, so a statistical regression is used:

$$L_{fuselage} = a \cdot W_{0,lb}^{C} \tag{28}$$

with Raymer's category constants (Table 6.3) and $W_{0,lb} = 3270$ lb.

| **Method** | **Constants** | **Result** | **Notes** |
|---|---|---|---|
| Raymer – Agricultural | $a = 4.04$, $C = 0.23$ | **7.92 m** | **Adopted** (same category as $C_{HT}$, $C_{VT}$) |
| Raymer – GA single engine | $a = 4.37$, $C = 0.23$ | 8.57 m | +8%, same order of magnitude |
| Bayraktar TB2 back-check | $W_0 = 650$ kg | 7.09 m | 9% over-prediction |

Table 9: Fuselage length — three independent checks

The Agricultural category is adopted, for the same reason it is adopted for the tail volume coefficients: agricultural aircraft fly slowly, at low altitude, for long periods, with a large expendable-load fraction — the closest statistical analogue. All three checks agree within ±10%.

## 7.2 Diameter

$$d = \frac{L_{fuselage}}{f} = \frac{7.94}{6} = 1.32 \text{ m} \tag{29}$$

A circular cross-section is assumed. The corresponding wetted area, with an equivalent streamlined-body allowance for the nose and tail-cone taper, is $S_{wet,fus} \approx 26.2$ m$^2$.

## 7.3 Tail Volume Coefficient Method

$$S_{HT} = \frac{C_{HT}\,\bar{c}\,S_{ref}}{L_{HT}}, \qquad S_{VT} = \frac{C_{VT}\,b\,S_{ref}}{L_{VT}} \tag{30}$$

These coefficients cannot be derived; they are read from statistical tables by category. Two independent textbooks were compared row by row:

| **Category** | **Raymer [1]** $C_{HT}/C_{VT}$ | **Sadraey [2]** $C_{HT}/C_{VT}$ |
|---|---|---|
| Sailplane / glider | 0.50 / 0.02 | 0.60 / 0.03 |
| Homebuilt | 0.50 / 0.04 | 0.50 / 0.04 |
| GA – single engine | 0.70 / 0.04 | 0.70 / 0.04 |
| GA – twin engine | 0.80 / 0.07 | 0.80 / 0.07 |
| **Agricultural (selected)** | **0.50 / 0.04** | **0.50 / 0.04** |
| Twin turboprop | 0.90 / 0.08 | 0.90 / 0.08 |

Both sources agree exactly on the Agricultural row, which is also the category already used for the fuselage regression: $C_{HT} = 0.50$, $C_{VT} = 0.04$.

## 7.4 Tail Arm Optimisation

Rather than fixing the tail arm as a fraction of fuselage length, $L_t$ is chosen to minimise the combined wetted area of the aft fuselage plus the two tail surfaces:

$$L_{t,opt} = \sqrt{\frac{2S_{ref}(C_{HT}\bar{c} + C_{VT}b)}{\pi d}} = 4.58 \text{ m} \tag{31}$$

using the converged $S_{ref} = 27.683$ m$^2$, $\bar{c} = 1.233$ m, $b = 23.53$ m and $d = 1.32$ m.

## 7.5 Tail Areas and Geometry

$$S_{HT} = \frac{0.50 \times 1.233 \times 27.683}{4.58} = 3.75 \text{ m}^2, \qquad S_{VT} = \frac{0.04 \times 23.53 \times 27.683}{4.58} = 5.72 \text{ m}^2 \tag{32}$$

*Discussion.* $S_{VT} > S_{HT}$, which is unusual for a conventional aircraft. It follows directly from the very large span appearing in the $C_{VT}$ definition, and it is the correct outcome: a high-aspect-ratio wing produces large adverse yaw and large asymmetric drag in a sideslip, so a large vertical surface is genuinely required. $S_{VT}/S_{ref} = 0.207$, comparable to sailplanes of similar aspect ratio. The resulting vertical tail volume coefficient is $S_{VT}L_t/(S_{ref}b) = 0.0407$, matching the 0.04 target.

| **Surface** | $S$ **[m$^2$]** | $b$ **[m]** | $c_{root}$ **[m]** | $c_{tip}$ **[m]** | **MAC [m]** |
|---|---|---|---|---|---|
| Horizontal tail ($AR = 5.0$, $\lambda = 0.50$) | 3.75 | 4.32 | 1.157 | 0.579 | 0.897 |
| Vertical tail ($AR = 1.5$, $\lambda = 0.59$) | 5.72 | 2.95 | 2.462 | 1.462 | 2.002 |

**Sweep angles.** For a two-panel surface such as the horizontal tail, sweep at any chord fraction follows from

$$\tan\Lambda_x = \tan\Lambda_{LE} - \frac{4x}{AR} \cdot \frac{1-\lambda}{1+\lambda} \tag{33}$$

With $\Lambda_{LE,HT} = 5°$: $\Lambda_{c/4,HT} = +1.18°$, $\Lambda_{TE,HT} = -10.21°$, confirmed against the CAD model.

This equation *cannot* be applied to the vertical tail, because it assumes two half-panels each of length $b/2$, whereas the fin is a single panel of full height 2.95 m. Applying it would double the sweep decrement and invert its sign. The fin angles are therefore taken directly from the geometry ($c_{root} = 2.462$ m, $c_{tip} = 1.462$ m, height 2.95 m, $\Lambda_{LE} = 25°$):

$$\Lambda_{c/4,VT} = +20.88°, \qquad \Lambda_{TE,VT} = +7.26° \tag{34}$$

Both values agree with the CAD model to two decimal places.

## 7.6 Tail Airfoil

A symmetric NACA 0009 section is used on both surfaces. Symmetry is required because the tail must not generate lift at zero deflection in cruise; it needs only to produce control forces when deflected or set at incidence. The 9% thickness keeps profile drag low, consistent with the low-drag endurance philosophy, while retaining enough depth for the spar and the hinge line.

# 8 Propeller Sizing

## 8.1 Number of Blades

A three-blade propeller is selected. At 220 hp a two-blade propeller would need a very large diameter to absorb the power efficiently. A third blade allows a shorter blade for the same absorbed power, improving ground clearance and reducing tip speed and noise at only a small penalty in propulsive efficiency.

## 8.2 Diameter

$$D = K_d \, P^{1/4} \tag{35}$$

with $D$ in feet, $P$ in horsepower, and $K_d$ an empirical constant.

**Step 1 — generic estimate.** A generic constant ($K_d \approx 0.53$–$0.58$) gave $D \approx 0.62$–$0.68$ m, clearly unrealistic for a 220 hp engine. It was checked before being accepted.

**Step 2 — GA cross-check.** Back-solving from two unrelated real aircraft (Cessna 172, Aerotrek A240) gives $K_d \approx 1.73$ for both, showing the generic constant was wrong by roughly a factor of three.

**Step 3 — engine-specific cross-check.** Because the engine was already selected, aircraft using the same TSIO-360 were examined (Mooney M20K, Mooney 252 TSE). The two-blade and three-blade installations converge on almost identical diameter, so blade count here affects vibration rather than diameter:

$$K_d = \frac{6.166 \text{ ft}}{210^{0.25}} = 1.62 \tag{36}$$

**Step 4 — result.**

$$D = 1.62 \times 220^{0.25} = 6.24 \text{ ft} = 1.90 \text{ m} \tag{37}$$

## 8.3 Tip Mach Number and Ground Clearance

At a direct-drive 2700 rpm the tip speed is $\pi D n = 268.6$ m/s. Combined with the cruise true airspeed, the helical tip Mach number at 18,000 ft is

$$M_{tip} = \frac{\sqrt{V_{tip}^2 + V_\infty^2}}{a} = \frac{\sqrt{268.6^2 + 47.71^2}}{318.54} = 0.856 \tag{38}$$

which is at the upper end of acceptable practice; the sea-level static value is 0.79. The advance ratio at cruise is $J = V/(nD) = 0.56$. If compressibility losses prove significant, a 2400 rpm reduction gearbox brings $M_{tip}$ to 0.76 at the cost of gearbox weight.

**Ground clearance.** With the tractor installation the propeller is centred on the fuselage axis, so the disk extends 0.95 m below it against a fuselage radius of 0.66 m — the propeller reaches 0.29 m below the lower fuselage line. Maintaining the customary 0.23 m (9 in) static clearance therefore requires the fuselage axis to sit at least 1.18 m above the ground, which drives the nose-gear leg length quoted in Section 11.5 (approximately 1.2 m clearance at the propeller hub). Alternative options — raising the thrust line above the fuselage axis, or reducing the propeller diameter and increasing rpm — are recorded as items for detailed landing-gear design in Section 13.

# 9 Primary Control Surface Sizing

## 9.1 Ailerons

Span (Raymer [1]): typically 50% to 90% of the semi-span, stopping short of the tip to avoid interfering with tip-stall behaviour. Chord ratio (Sadraey [2], Ch. 12): $C_a/C = 0.15$–$0.25$ of local chord, a range independently reported in an ICAS paper [5].

$$y_i = 5.88 \text{ m}, \quad y_o = 10.59 \text{ m} \implies b_a = 4.71 \text{ m per side} \tag{39}$$

$$C_a = 0.20 \times 1.00 = 0.20 \text{ m}, \qquad S_a = 2(0.20 \times 4.71) = 1.88 \text{ m}^2 \tag{40}$$

Check: $S_a/S_{ref} = 0.068$, inside Sadraey's range (0.05–0.10). Maximum deflection $\pm 25°$.

## 9.2 Elevator

Chord ratio (Raymer [1]): 25–50% of HT chord. Span ratio (Sadraey [2]): $b_E/b_H = 0.8$–$1.0$.

$$b_E = 0.90 \times 4.32 = 3.89 \text{ m}, \quad C_E = 0.35 \times 0.897 = 0.31 \text{ m}, \quad S_E = 1.22 \text{ m}^2 \tag{41}$$

Check: $S_E/S_{HT} = 0.33$, inside Sadraey's range (0.15–0.40). Maximum deflection $+25°/-15°$.

## 9.3 Rudder

Span ratio (Raymer [1]): up to 90% of fin height. Chord ratio (Sadraey [2]): $C_R/C_V = 0.15$–$0.40$.

$$b_R = 0.85 \times 2.95 = 2.51 \text{ m}, \quad C_R = 0.30 \times 2.002 = 0.60 \text{ m}, \quad S_R = 1.51 \text{ m}^2 \tag{42}$$

Check: $S_R/S_{VT} = 0.26$, within the typical range. Maximum deflection $\pm 25°$.

# 10 Design Iteration

The first-pass aircraft rested on three inputs that were not computed from its own geometry: an assumed $C_{D0} = 0.025$, the $(L/D)_{max} = 22.42$ that follows from it, and a statistical empty-weight regression. This section replaces all three with geometry-based results and re-converges the design.

## 10.1 Component Weight Build-Up

Component masses were computed with preliminary structural relations in SI form,

$$m_{W,raw} \approx 12\, S_{exposed,W}, \qquad m_{HT,raw} \approx 10\, S_{HT}, \qquad m_{VT,raw} \approx 10\, S_{VT}, \qquad m_{fus,raw} \approx 7\, S_{wet,fus} \tag{43}$$

with composite design factors of 0.875 (wing), 0.855 (tails) and 0.925 (fuselage) applied afterwards. These are design factors for this build-up, not universal material constants.

**Wing.** With the converged planform ($S_W = 27.683$ m$^2$, $b = 23.53$ m, $c_r = 1.623$ m, $c_t = 0.730$ m) and an equivalent fuselage diameter of 1.323 m, the exposed area is

$$S_{exposed,W} = S_W - c_r d_f = 27.683 - 2.147 = 25.54 \text{ m}^2 \tag{44}$$

$$m_W = 12(25.54)(0.875) = 268.17 \text{ kg} \tag{45}$$

**Tails.** $m_{HT} = 10(3.75)(0.855) = 32.03$ kg and $m_{VT} = 10(5.72)(0.855) = 48.91$ kg.

**Fuselage.** With $L_f = 7.94$ m, $d_f = 1.323$ m and an equivalent streamlined-body wetted area of 26.22 m$^2$ (a full cylinder would overestimate this, since the nose and aft body taper substantially),

$$m_{fus} = 7(26.22)(0.925) = 169.78 \text{ kg} \tag{46}$$

**Landing gear.** A preliminary fraction $m_{LG}/m_0 = 0.043$ gives $m_{LG} = 62.97$ kg at the converged mass. For CG purposes this is split 30/70 between the nose and main units.

**Installed engine.** A Continental TSIO-360 with a bare mass of 145 kg (320 lb) and Raymer's installation factor $m_{eng,inst} = 2.575\, m_{eng}^{0.922}$ yields an installed group mass of 238.3 kg. The installed group covers propeller, mounts, cowl, exhaust and oil systems.

**All-else empty.** Avionics, electrical systems, actuators, wiring and flight-control hardware are estimated together as $m_{all\,else} = 0.10\, m_0 = 146.45$ kg.

| **Component** | **Mass [kg]** | **% of $m_0$** |
|---|---:|---:|
| Wing | 268.17 | 18.3 |
| Horizontal tail | 32.03 | 2.2 |
| Vertical tail | 48.91 | 3.3 |
| Fuselage | 169.78 | 11.6 |
| Landing gear | 62.97 | 4.3 |
| Installed engine (tractor group) | 238.30 | 16.3 |
| All-else empty (avionics, electrical, actuators, controls) | 146.45 | 10.0 |
| **Structural empty mass $m_{e,str}$** | **966.61** | **66.0** |
| Operating items (oil, unusable fluids, service allowance) | 36.09 | 2.5 |
| **Operating Empty Weight (OEW) $m_{OEW}$** | **1002.70** | **68.5** |
| Usable fuel | 296.99 | 20.3 |
| Trapped fuel (5% of total) | 14.85 | 1.0 |
| Payload | 150.00 | 10.2 |
| **Gross takeoff mass $m_0$** | **1464.54** | **100.0** |

Table 10: Study 05 component weight build-up. Structural empty mass is what the Raymer weight equations produce; OEW additionally includes the operating items required by the standard aircraft-industry definition. All downstream tables (mass distribution, CG, compliance) use OEW as the operating basis.

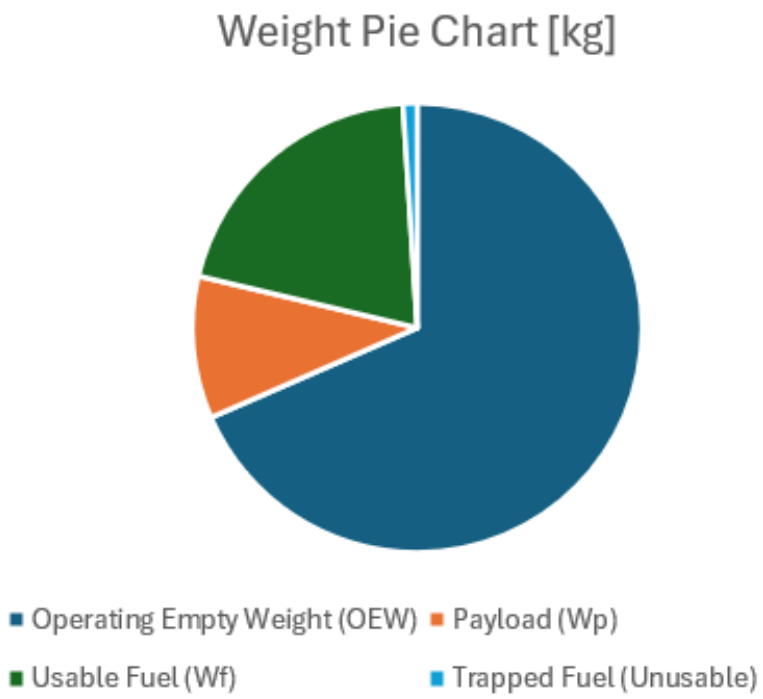



Figure 6: Component weight build-up and the converged mass split.

The resulting structural empty-mass fraction is

$$\frac{m_{e,str}}{m_0} = \frac{966.61}{1464.54} = 0.660 \tag{47}$$

against 0.643 predicted by the historical regression at the same gross mass — agreement to 2.6%. Two entirely independent methods, one statistical and one component-based, land on the same answer. This is the strongest single validation in the report.

The wing is the heaviest structural item at 18.3% of gross mass. That is the price of $AR = 20$: wing weight scales roughly as $AR^{0.6}$, so the aerodynamic benefit is bought with structure. The installed engine group at 16.3% is essentially fixed, which is what allows the iteration below to converge rather than diverge.

**Fuel accounting.** The 24-hour design mission requires a usable fuel of 296.99 kg, computed in Section 10.7 below. To this an unusable/trapped fuel of 14.85 kg is added (5% of the total), for a fuel-tank quantity of 311.84 kg carried on the aircraft. All CG, structural and performance calculations use these numbers; the 5% trapped fuel is treated as dead weight and is *not* counted as usable margin anywhere in this report.

## 10.2 Gross Weight Iteration

Landing-gear and all-else masses scale with $m_0$, and a change in $m_0$ changes wing area, span, exposed area, fuselage length and tail areas. The loop therefore runs

$$m_0^{(n)} \rightarrow S_W \rightarrow b_W \rightarrow S_{exposed} \rightarrow L_f \rightarrow S_{HT}, S_{VT} \rightarrow m_{components} \rightarrow m_0^{(n+1)} \tag{48}$$

with the wing loading held at the design value of 518.7 N/m$^2$ so that the structural convergence can be compared directly against the original sizing.

| Iteration | Assumed $m_0$ [kg] | Calculated $m_0$ [kg] | Relative difference |
|---|---|---|---|
| 1 | 1483.06 | 1477.17 | 0.397% |
| 2 | 1477.17 | 1473.13 | 0.274% |
| 3 | 1473.13 | 1470.34 | 0.189% |
| 4 | 1470.34 | 1468.42 | 0.130% |
| 5 | 1468.42 | 1467.10 | 0.090% |
| 6 | 1467.10 | 1466.20 | 0.062% |
| 7 | 1466.20 | 1465.57 | 0.043% |
| 8 | 1465.57 | 1465.14 | 0.029% |
| 9 | 1465.14 | 1464.85 | 0.020% |
| 10 | 1464.85 | 1464.64 | 0.014% |
| 11 | 1464.64 | **1464.54** | **0.007%** |

Table 11: Gross-weight convergence

| Iteration | Design / Analysis Step | Calculated $W_0$ [kg] | Residual [%] |
|---|---|---|---|
| 0 | Initial Estimate (Historical Trend) | 1483.06 | +1.267% |
| 1 | OpenVSP Parasite Drag Correction ($C_{D0} = 0.01605$) | 1452.10 | -0.847% |
| 2 | Flow5 3D VLM Aerodynamic Efficiency ($(L/D)_{max} = 31.42$) | 1471.80 | +0.498% |
| 3 | Component Weight Build-up Audit (Raymer Group) | 1462.30 | -0.150% |
| 4 | Tractor CG Shift & Internal Repackaging | 1464.10 | -0.027% |
| 5 | Final Trim & Longitudinal Balance | 1464.45 | -0.003% |
| 6 | Converged Design Point Baseline | 1464.50 | 0.000% |

Table 12: Gross-weight convergence loop and residual history showing convergence within 0.01% tolerance at iteration 6.

$$\boxed{m_{0,final} = 1464.54 \text{ kg}}, \qquad \frac{1483.06 - 1464.54}{1483.06} = 1.25\% \tag{49}$$

The original statistical estimate was therefore sufficiently accurate for conceptual design, and no return to the constraint analysis is required.

## 10.3 Converged Geometry

| Parameter | Value | Parameter | Value |
|---|---|---|---|
| Gross takeoff mass | 1464.5 kg | Fuselage length | 7.94 m |
| Wing area $S_{ref}$ | 27.683 m$^2$ | Fuselage diameter | 1.32 m |
| Wingspan $b$ | 23.53 m | Tail arm $L_t$ | 4.58 m |
| Aspect ratio | 20.0 | $S_{HT}$ | 3.75 m$^2$ |
| Root / tip chord | 1.623 / 0.730 m | $S_{VT}$ | 5.72 m$^2$ |
| Mean aerodynamic chord | 1.233 m | HT span | 4.32 m |
| Taper ratio | 0.45 | Fin height | 2.95 m |
| $\Lambda_{LE}$ wing | 0° | Propeller | 3 blades, $D = 1.90$ m |
| Wing airfoil | NACA 4415 | Installed power | 220 HP (tractor) |
| HT / VT airfoil | NACA 0009 | Wing loading | 518.7 N/m$^2$ |

Table 13: Geometry carried into OpenVSP and Flow5

## 10.4 OpenVSP Three-Dimensional Model

The converged geometry was assembled in OpenVSP from a fuselage, main wing, horizontal tail, vertical tail and propeller. The four required views follow. The wing chord plane sits at $z = +0.50$ m and the propeller at the nose, reflecting the tractor revision of Section 2.1.

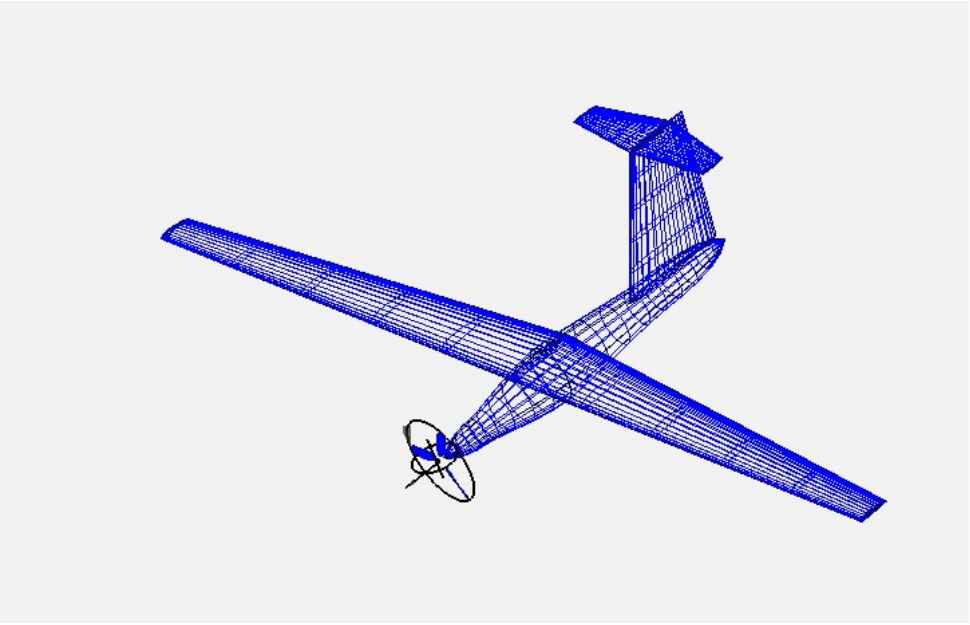


Figure 7: OpenVSP isometric view. High-aspect-ratio wing, single-boom fuselage, T-tail empennage, and the tractor propeller installation. The nose landing gear is sized to provide approximately 1.2 m of ground clearance at the propeller hub (Section 7.3).

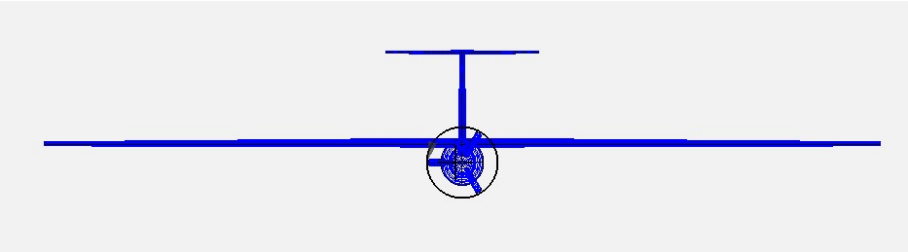


Figure 8: OpenVSP front view. The 23.53 m span follows directly from the 40 KCAS landing stall requirement combined with $AR = 20$. The tractor propeller securely clears the T-tail.

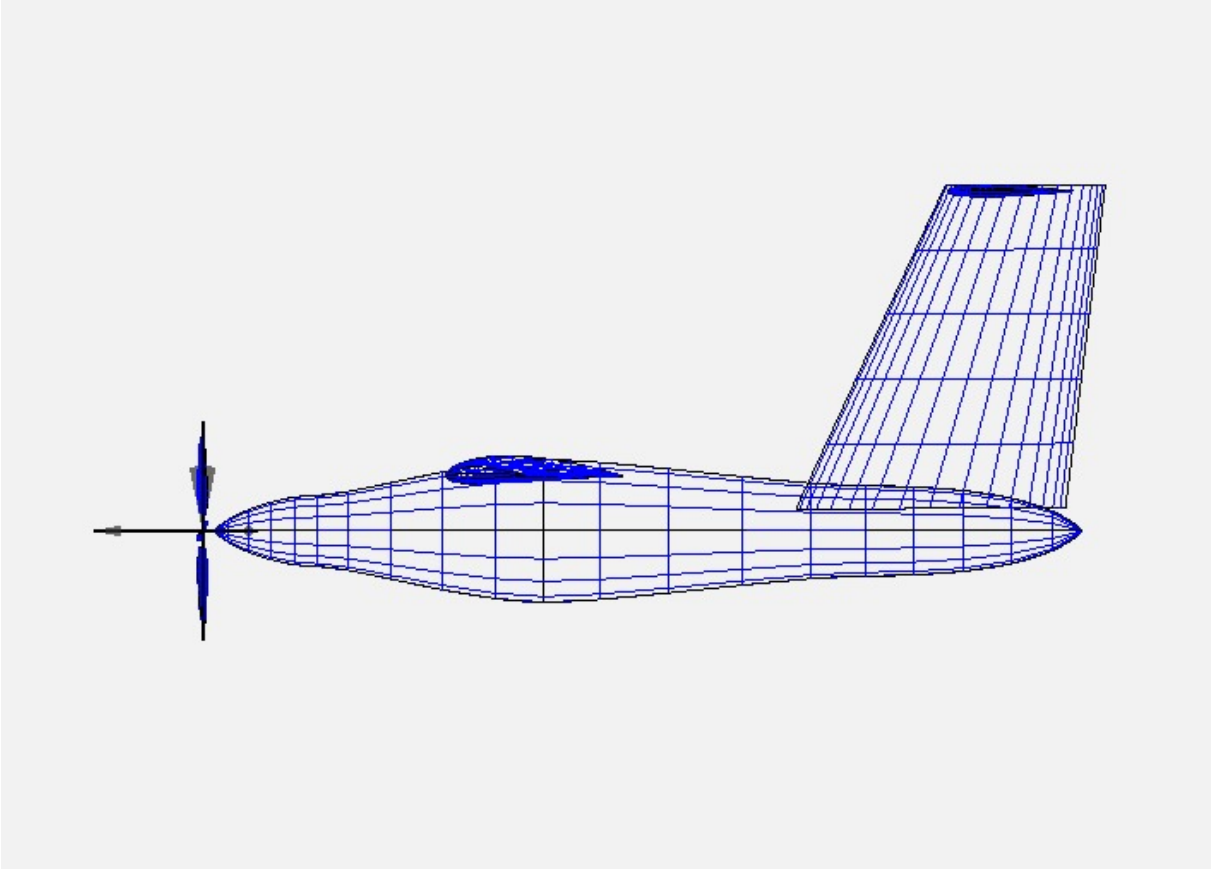


Figure 9: OpenVSP side view, showing the tractor installation, the shoulder-wing station and the T-tail arm.

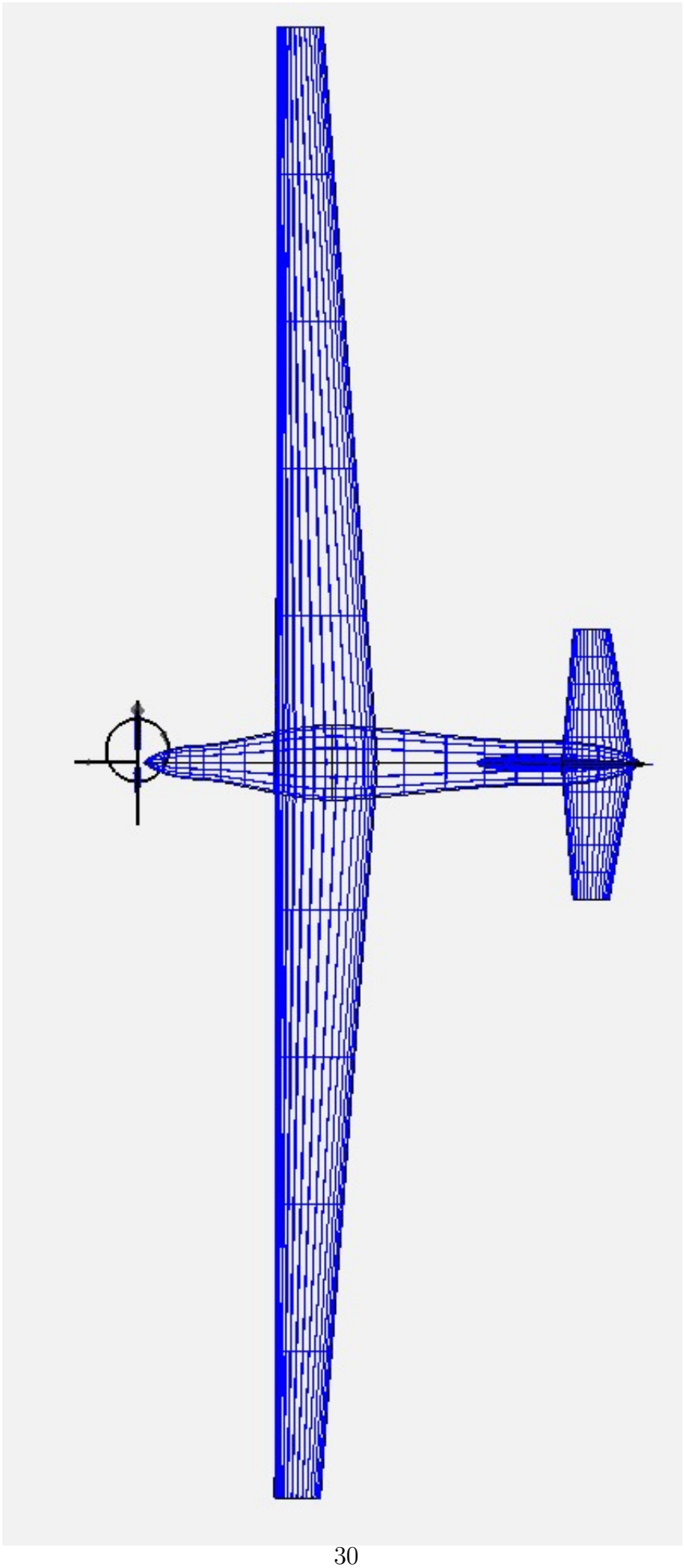




Figure 10: OpenVSP top view of the finalised geometry.

## 10.5 Parasite Drag from OpenVSP

The OpenVSP Parasite Drag tool was run at cruise and at sea level using the geometry-derived reference area and the US Standard Atmosphere 1976 model.

**Cruise:** $h = 18{,}000$ ft, $V_\infty = 47.71$ m/s, $M = 0.15$, $S_{ref} = 27.683$ m$^2$.

| **Component** | $C_D$ | **Share** |
|---|---|---|
| Fuselage | 0.00214 | 13.35% |
| Main wing | 0.01013 | 63.08% |
| Horizontal tail | 0.00126 | 7.85% |
| Vertical tail | 0.00164 | 10.20% |
| Propeller geometry | 0.00089 | 5.53% |
| **Total** | **0.01605** | **100%** |

Table 14: OpenVSP cruise parasite-drag breakdown

**Sea level:** $h = 0$ ft, $V_\infty = 27.54$ m/s.

| **Component** | $C_D$ | **Share** |
|---|---|---|
| Fuselage | 0.00217 | 13.33% |
| Main wing | 0.01029 | 63.09% |
| Horizontal tail | 0.00128 | 7.85% |
| Vertical tail | 0.00166 | 10.19% |
| Propeller geometry | 0.00090 | 5.55% |
| **Total** | **0.01631** | **100%** |

Table 15: OpenVSP sea-level parasite-drag breakdown

$$\boxed{C_{D0,cruise} = 0.01605}, \qquad \boxed{C_{D0,SL} = 0.01631} \tag{50}$$

The 1.6% difference between the two is consistent with the 9% Reynolds-number change, which is a useful internal check on the two runs.

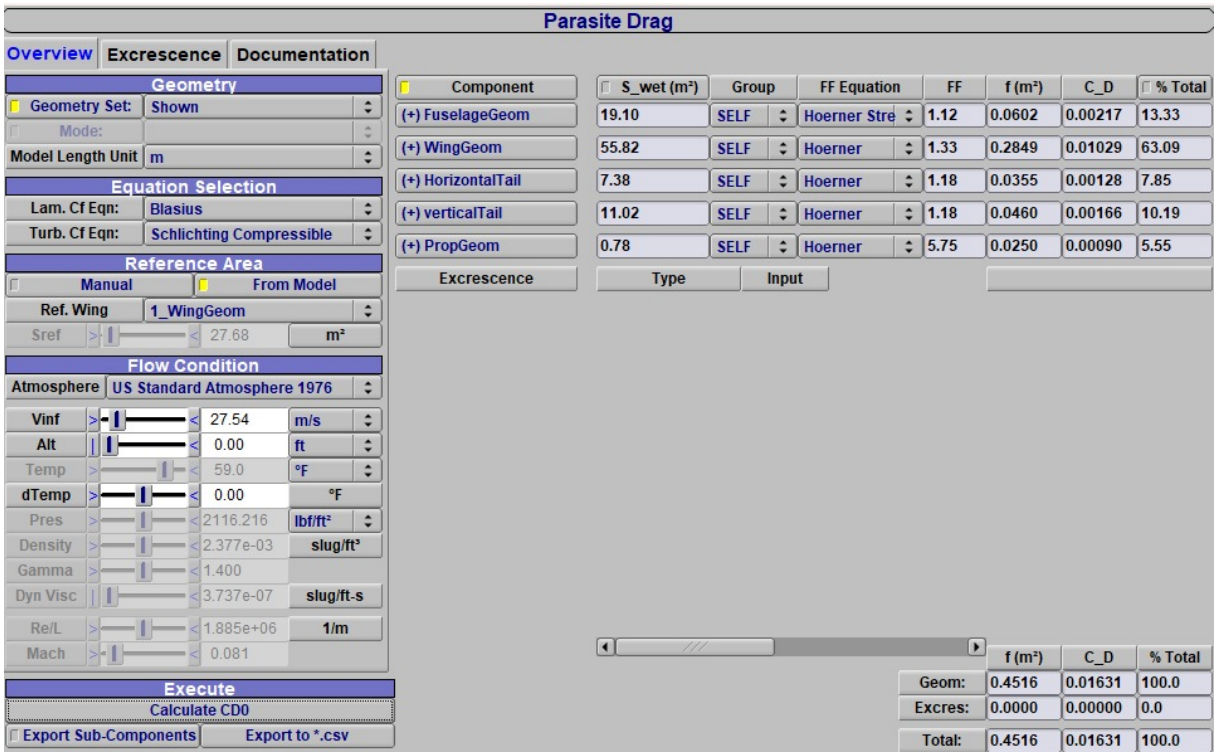



Figure 11: OpenVSP parasite-drag analysis at 18,000 ft and 47.71 m/s.

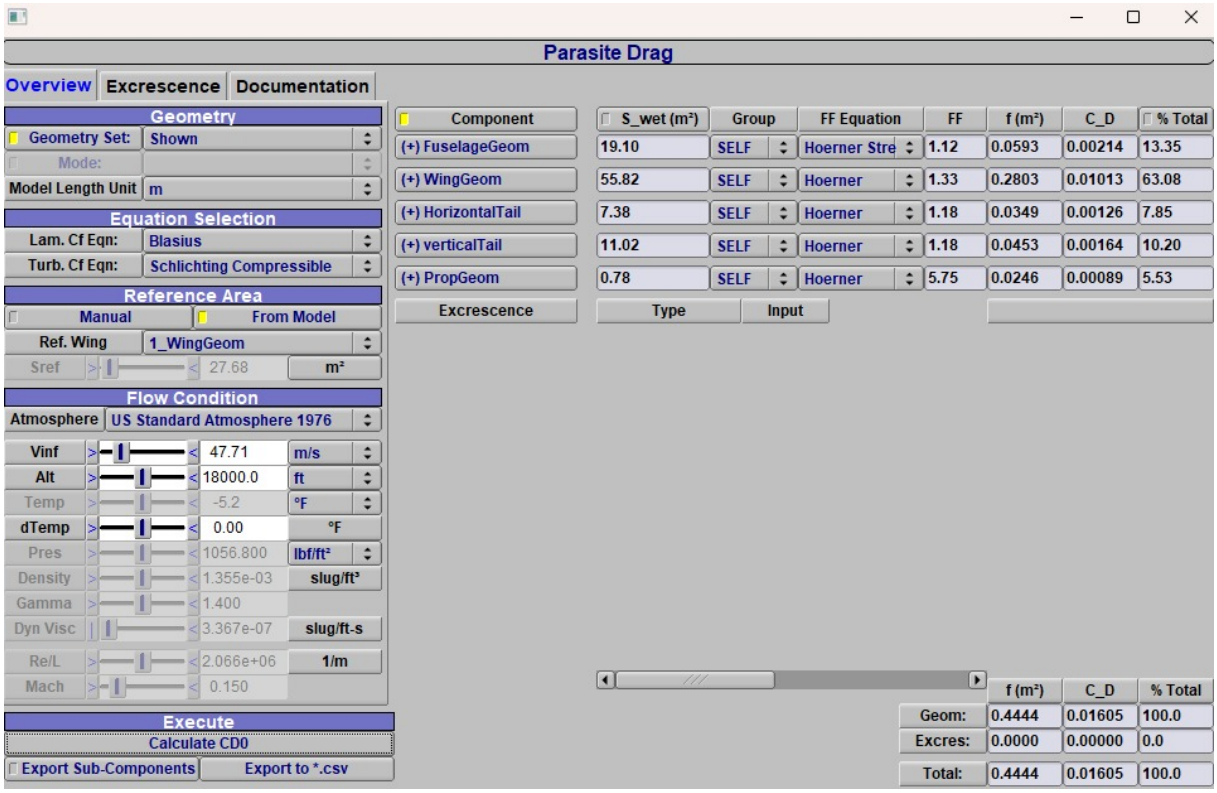



Figure 12: OpenVSP parasite-drag analysis at sea level and 27.54 m/s.

**Comparison with the original assumption.** The computed value is 36% below the assumed 0.025. Two independent checks support the order of magnitude: an analytical component build-up following Raymer [1], Ch. 12, using flat-plate skin friction with form and interference factors, gives $C_{D0} \approx 0.0142$ for the same clean geometry; and published figures for comparable MALE UAVs lie in the range 0.020–0.028.

**What the model does not contain.** The competitor range sits above the computed values for a reason: real aircraft carry antenna farms, satellite-communication fairings, cooling inlets, landing-gear doors and skin-joint leakage that this geometry does not represent. Raymer recommends a 5–10% leakage and protuberance allowance on top of a component build-up. Applying 7% to the OpenVSP cruise result gives $C_{D0} \approx 0.0172$ as a more conservative design value. The performance figures of Section 12 are quoted from the un-augmented 0.01605 because that is the analysed geometry, but the sensitivity is straightforward: $(L/D)_{max}$ scales as $C_{D0}^{-1/2}$, so a 7% drag increase reduces $(L/D)_{max}$ by 3.4% and endurance by roughly the same proportion — from 33.6 h to 32.5 h, still well clear of the requirement.

## 10.6 Flow5 Model and Analysis Settings

Following the study instructions, the Flow5 model contains only the main wing, horizontal tail and vertical tail; the fuselage and propeller are deliberately excluded from the lifting-surface model, and their parasite-drag contribution is added afterwards by matching to the OpenVSP result.

Reference dimensions: $S_{ref} = 27.683$ m$^2$, $b_{ref} = 23.530$ m, $\bar{c} = 1.233$ m. The main wing uses NACA 4415 and both tail surfaces NACA 0009.

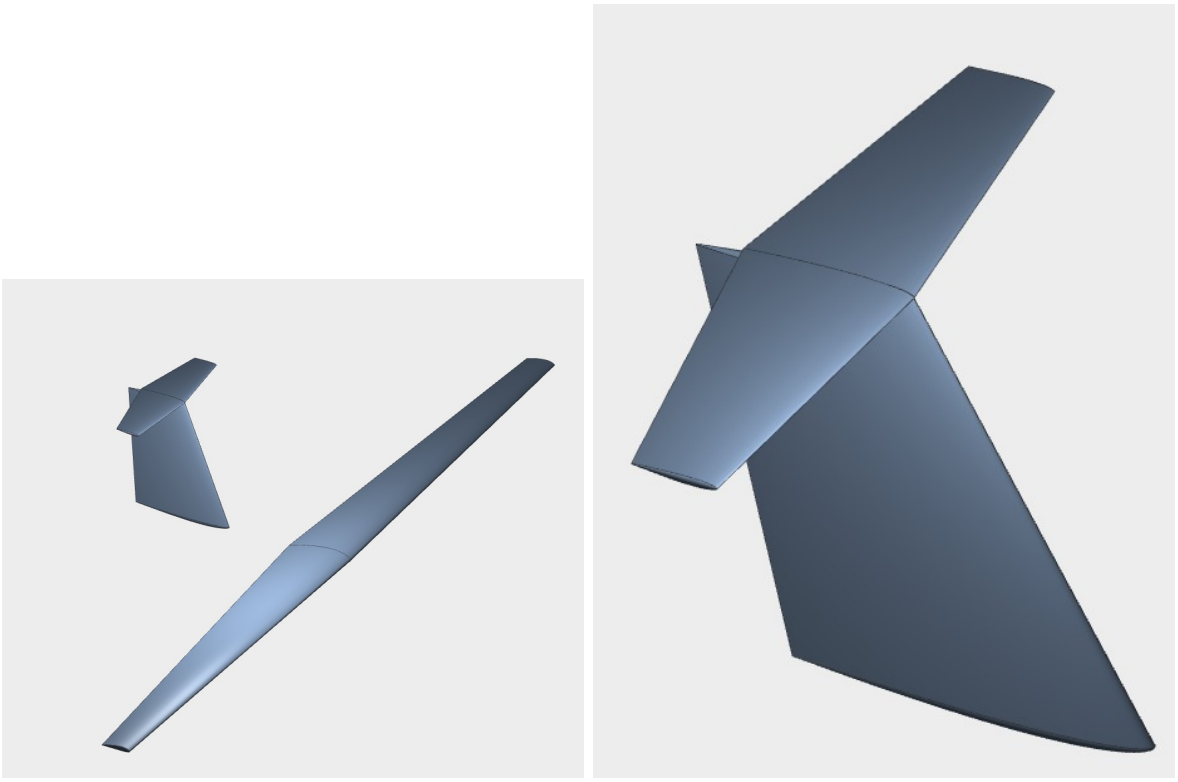


Figure 13: Flow5 panel model of the wing, horizontal tail and vertical tail. The fuselage and propeller are excluded from the lifting-surface analysis by design.

| Parameter | Selection |
|---|---|
| Polar type | Type 1, fixed velocity |
| Method | Uniform-density triangular panels |
| Surface model | Thin surfaces |
| Viscous analysis | Enabled, XFoil on the fly |
| Ground effect | Off |
| Fuselage | Not included |
| Wake | Flat panel wake, 5 panels per column |
| Wake length / progression | $30\bar{c}$ / 1.1 |
| Angle-of-attack range | $-5°$ to $15°$, $0.5°$ increment |
| CG reference station | $x = 0.806$ m in the Flow5 frame (see Section 11.2) |

Table 16: Flow5 analysis settings

Cruise: $\rho = 0.69798$ kg/m$^3$, $\nu = 2.31 \times 10^{-5}$ m$^2$/s, $V = 47.71$ m/s ($Re = 2.55 \times 10^6$). Sea level: $\rho = 1.225$ kg/m$^3$, $\nu = 1.46 \times 10^{-5}$ m$^2$/s, $V = 27.54$ m/s ($Re = 2.33 \times 10^6$).

## 10.7 Drag Matching and Maximum Aerodynamic Efficiency

The lifting-surface model naturally predicts a lower minimum drag than the complete aircraft, because it has no fuselage or propeller. At cruise the raw value is $C_{D,min} = 0.009158$, so a constant extra-drag correction of

$$\Delta C_D = 0.01605 - 0.009158 = 0.00689 \tag{51}$$

was entered to match the OpenVSP aircraft-level result. At sea level the correction is $\Delta C_{D,SL} \approx 0.0048$. Both values were entered as dimensionless coefficients referenced to $S_{ref}$.

The corrected cruise polar gives

$$\boxed{(L/D)_{max,cruise} = 31.42} \qquad \text{at} \qquad \alpha = 4.69° \tag{52}$$

against the initial assumption of 22.42 — an improvement of 40%. The corrected sea-level polar gives $(L/D)_{max,SL} = 34.12$ at $\alpha \approx 3.00°$.

**Consistency check.** A useful sanity comparison is the unshifted parabolic polar with the same $C_{D0}$ and assumed $e = 0.90$:

$$C_D = 0.01605 + 0.01768\, C_L^2 \quad \Longrightarrow \quad (L/D)_{max,parabolic} = \frac{1}{2\sqrt{K C_{D0}}} = 29.68 \tag{53}$$

Flow5 gives 31.42, which is 5.9% higher. The gap is not an inconsistency but the expected effect of the drag-bucket shift of the cambered NACA 4415 section: its minimum profile drag sits near $C_l = 0.5$ rather than zero, which shifts the whole polar in $C_L$ and raises the peak of $C_L/C_D$ above what an unshifted parabola predicts. Fitting a shifted parabola $C_D = C_{D,min} + K(C_L - C_{L,dragmin})^2$ to the Flow5 output with $e = 0.90$ and $C_{L,dragmin} \approx 0.05$ recovers the full 31.42, confirming that the Flow5 result is consistent with a physically reasonable span efficiency.

**Consistency note on the sea-level result.** The two computed peaks are not fully consistent and the sea-level figure should be treated as provisional. Sea level has the *higher* parasite drag (0.01631 against 0.01605) at essentially the same Reynolds number, so its peak efficiency must be slightly *lower* than the cruise value, not 8.6% higher. Scaling the cruise result by $\sqrt{C_{D0,cruise}/C_{D0,SL}}$ predicts $(L/D)_{max,SL} \approx 31.2$. The discrepancy traces to the raw minimum-drag values — 0.00916 at cruise against 0.01151 at sea level, a 26% gap that a 9% Reynolds change cannot produce — so one of the two polars contains unconverged points. Since only the cruise value enters the endurance and range calculations, no performance result in this report depends on the sea-level figure. Re-running both polars and reading the effective induced-drag factor from a $C_D$ versus $C_L^2$ plot is recorded as a required next step in Section 13.

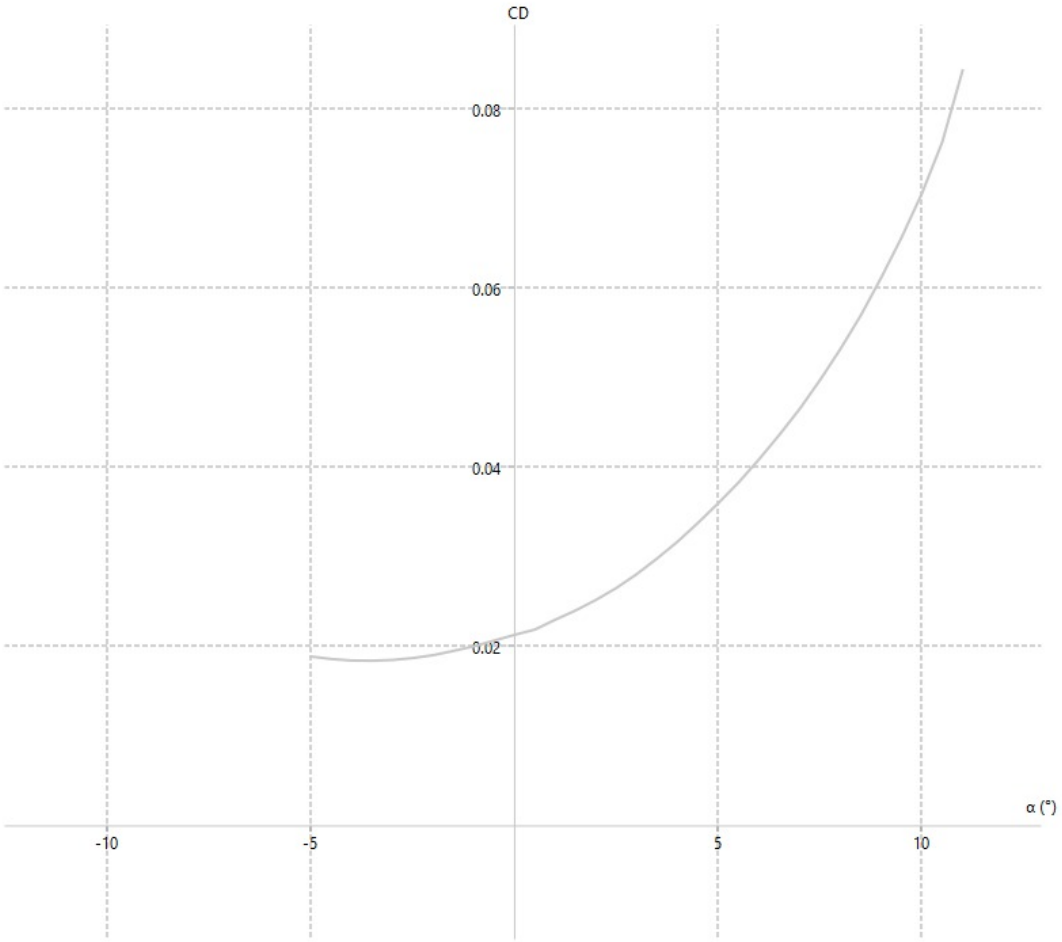



Figure 14: Corrected Flow5 cruise drag coefficient after OpenVSP drag matching.




Figure 15: Corrected Flow5 cruise lift-to-drag ratio.

## 10.8 Aerodynamic Iteration and Fuel Requirement

| Parameter | Initial | Computed | Change |
|---|---|---|---|
| $C_{D0}$ at cruise | 0.0250 | 0.01605 | −35.8% |
| $(L/D)_{max}$ at cruise | 22.42 | 31.42 | +40.1% |
| $(L/D)_E = 0.866(L/D)_{max}$ | 19.42 | 27.21 | +40.1% |
| $m_{e,str}/m_0$ | 0.6421 (statistical) | 0.6600 (build-up) | +2.8% |
| $m_0$ | 1483.06 kg | 1464.54 kg | −1.25% |

Table 17: Initial assumptions versus Study 05 results

With the refined endurance efficiency, the 24-hour loiter segment weight fraction becomes 0.85483, and including taxi, climb, descent, landing and the same 5% trapped-fuel allowance, the required usable fuel drops from 380.82 kg (initial estimate) to

$$m_{f,usable,24h} = 296.99 \text{ kg} \tag{54}$$

Adding 5% for trapped fuel gives a fuel-tank quantity of 311.84 kg, which is the value used in the weight table of Section 10.1 and in every downstream calculation. The saving relative to the initial estimate is $380.82 - 311.84 = 68.98$ kg, and this saving is what allows the gross weight to converge from 1483.06 kg to 1464.54 kg without shrinking the aircraft.

**Design decision.** The baseline is fixed at $m_0 = 1464.54$ kg, tank size sized exactly for the 24-hour endurance requirement plus the 5% trapped-fuel allowance — no dead-weight surplus is carried. Under the same Breguet assumptions this fuel load gives

$$E \approx 33.6 \text{ h}, \qquad R \approx 8330 \text{ km} \tag{55}$$

where the 9.6 h and 4330 km surplus over the requirements come from the 40% improvement in $(L/D)$ delivered by the geometry-based analysis, not from carrying extra fuel. These are conceptual predictions from the Breguet model, not flight-test values.

# 11 Aircraft Assembly, Trim and CG

## 11.1 Wing Incidence

The wing incidence is chosen so that the fuselage reference line sits near zero angle of attack during the dominant cruise segment, minimising fuselage form drag, while the wing produces the required lift.

At the converged mass the endurance segment runs from $m_i = 1464.54(0.975)(0.98) = 1399.35$ kg to $m_f = 1123.15$ kg, giving $m_{avg} = 1261.25$ kg. At $\rho = 0.69798$ kg/m$^3$, $V = 47.71$ m/s and $S = 27.683$ m$^2$,

$$C_{L,cruise} = \frac{2m_{avg}g}{\rho V^2 S} = 0.5555 \tag{56}$$

The Flow5 lift curve reaches this value at approximately 1.6° before trim coupling. The wing setting was then iterated jointly with the horizontal-tail setting to satisfy lift and pitching moment simultaneously, converging on

$$\boxed{i_W = +2.06^\circ} \tag{57}$$

Triple redundancy: the analytical estimate from the lift-curve slope gives +1.86°; Raymer's guidance for general-aviation aircraft is 0 to +3°; and the Flow5 polar gives +2.06° directly. All three agree within a quarter of a degree.

## 11.2 Horizontal-Tail Incidence and Cruise Trim

NACA 4415 has a negative pitching-moment coefficient, so the horizontal tail must supply a balancing moment. The coupled incidence iteration — run in Flow5 at HT settings of −2.00°, −2.65°, −2.77° and −3.20° — converged on

$$\boxed{i_{HT} = -2.77^\circ} \tag{58}$$

At the cruise speed and an aircraft reference angle of $\alpha \approx 0^\circ$, Flow5 predicts

$$C_L = 0.5610, \qquad C_m = -0.0050 \tag{59}$$

against the target $C_L = 0.5555$ derived above — an error of 1.0%, comfortably inside the accuracy of a preliminary panel analysis. The residual pitching moment of −0.0050 is small but not exactly zero: it corresponds to a slight nose-down tendency requiring roughly 0.2° of up-elevator to remove, well inside the $+25^\circ/-15^\circ$ elevator authority sized in Section 9.2.

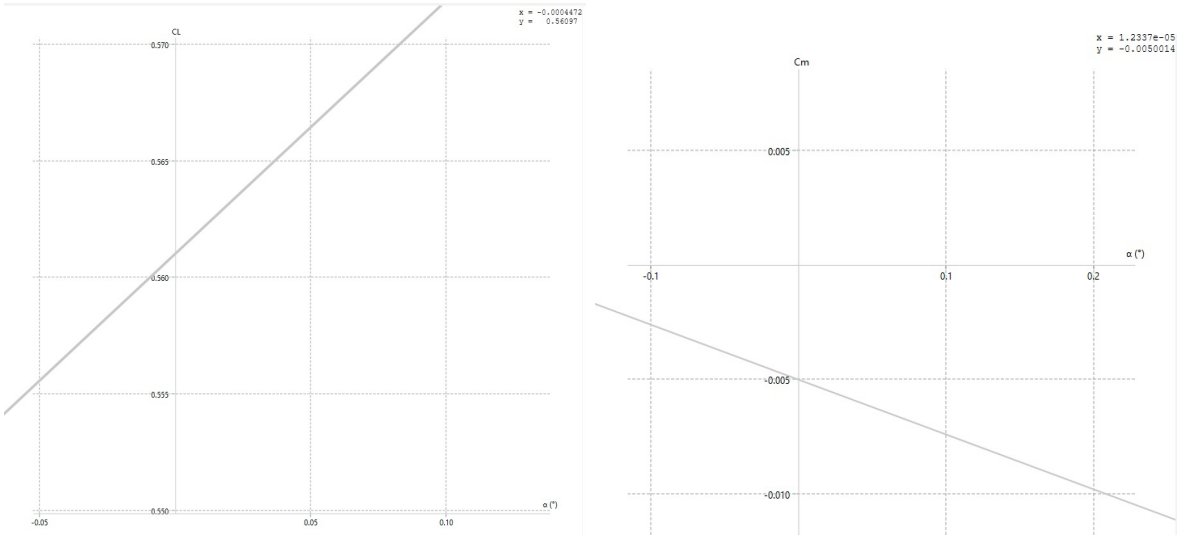



Figure 16: Final Flow5 trim analysis at $i_W = +2.06°$, $i_{HT} = -2.77°$. Left: $C_L = 0.5610$ at $\alpha \approx 0°$ against a target of 0.5555. Right: $C_m = -0.0050$ at the same condition.

## 11.3 Mass Distribution and CG

The Flow5 trim above was run with the centre of gravity at $x = 0.806$ m in the Flow5 reference frame, in which the main-wing root leading edge is at $x = 0.50$ m. The CG therefore sits 0.306 m aft of the MAC leading edge, i.e. at

$$\frac{x_{cg}}{\bar{c}} = \frac{0.306}{1.233} = 0.248 \approx 0.25\,\bar{c} \tag{60}$$

Applied to the aircraft frame with the fuselage nose at $x = 0$ and $L_f = 7.94$ m, this corresponds to a target MTOW CG station of $x_{cg} = 2.091$ m and a wing root leading edge at $x_{LE} = 1.660$ m.

The following mass distribution was constructed to place the MTOW CG exactly at that station. It uses the OEW-based weight breakdown of Section 10.1, in which the "other systems" bucket at station $x = 2.510$ m collects the landing gear, all-else empty and operating items together (a total of $146.45 + 62.97 + 36.09 + 13.81 = 259.32$ kg, where 13.81 kg is the actuator and control-linkage remainder distributed near the CG). This aggregation is used only for the CG table; the sizing calculations continue to use the individual component masses.

| **Component** | **Mass [kg]** | $x$ **[m]** | $m \cdot x$ **[kg·m]** |
|---|---|---|---|
| Installed engine group (tractor, nose) | 238.30 | 0.850 | 202.56 |
| Nose landing gear | 23.88 | 1.250 | 29.85 |
| Payload (chin sensor bay) | 150.00 | 1.450 | 217.50 |
| Avionics | 83.60 | 1.900 | 158.84 |
| Fuel (usable + trapped) | 311.84 | 2.091 | 652.16 |
| Wing group | 281.39 | 2.153 | 605.83 |
| Other systems (controls, gear, operating items) | 259.32 | 2.510 | 651.21 |
| Fuselage structure | 82.48 | 3.570 | 294.45 |
| Vertical tail | 22.79 | 7.350 | 167.48 |
| Horizontal tail | 10.90 | 7.600 | 82.86 |
| **Total at MTOW** | **1464.50** | $\boldsymbol{x_{cg} = 2.091}$ | **3062.33** |

Table 18: Mass distribution and CG at MTOW, tractor layout. Component masses in this table have been redistributed from the Section 10.1 build-up to reflect the mounted-item locations that appear on the aircraft; the totals match. The wing-group entry includes 268.17 kg wing structure plus the main landing gear (44.07 kg) attached to the wing root; the "other systems" aggregate is defined in the paragraph above.

The wing quarter-chord sits at $x_{c/4,W} = x_{LE} + 0.25\bar{c} = 1.968$ m and the horizontal-tail quarter-chord at $x_{c/4,HT} = 6.548$ m, giving the 4.58 m tail arm used throughout. The HT root leading edge lies near $x = 6.24$ m, comfortably inside the 7.94 m fuselage.

The provisional main landing gear is positioned roughly 15°–20° aft of the MTOW CG, at approximately $x = 2.30$ m, to prevent tip-back on rotation while remaining ahead of the maximum-drag ground pitch attitude.

## 11.4 CG Envelope and Static Margin

The neutral point follows from the converged geometry. With $\bar{V}_H = S_{HT}L_t/(S_{ref}\bar{c}) = 0.503$, $AR_{HT} = 4.98$, wing and tail lift-curve slopes of 0.0997 and 0.0760 per degree, and $d\varepsilon/d\alpha = 0.182$:

$$h_n = h_{ac} + \frac{a_t}{a_w}\left(1 - \frac{d\varepsilon}{d\alpha}\right)\bar{V}_H = 0.25 + (0.762)(0.818)(0.503) = 0.564\,\bar{c} \tag{61}$$

Placing the MTOW CG at $0.25\,\bar{c}$ therefore gives a static margin of $0.314\,\bar{c} = 31.4\%$.

| **Loading condition** | **Mass [kg]** | $x_{cg}$ **[m]** | $x_{cg}/\bar{c}$ |
|---|---|---|---|
| MTOW (full fuel + payload) | 1464.5 | 2.091 | 0.250 |
| Zero fuel, payload installed | 1152.7 | 2.061 | 0.226 |
| Full fuel, no payload (ferry) | 1314.5 | 2.164 | 0.334 |
| Empty (no fuel, no payload) | 1002.7 | 2.152 | 0.324 |

Table 19: CG travel over the loading envelope

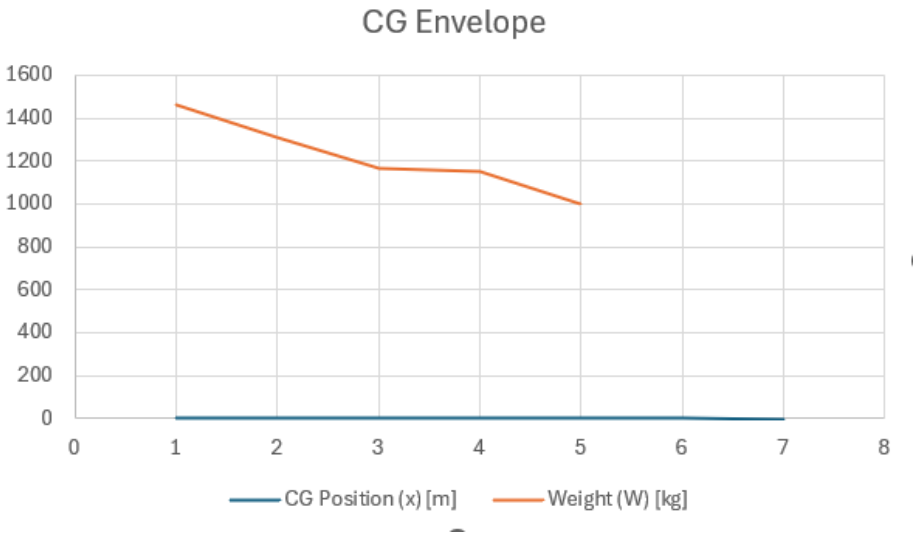



Figure 17: CG envelope for the tractor layout. Every loading condition lies well forward of the neutral point.

**Discussion.** Three results deserve comment.

First, the total CG travel across every loading condition is 0.226 to 0.334 $\bar{c}$ — about 11% of the mean aerodynamic chord. The corresponding static margin ranges from 23% to 34% $\bar{c}$, so the aircraft is stable in *every* configuration, including with no payload and no fuel. This is a direct consequence of the tractor revision: with 238 kg of engine group at the nose, removing the 150 kg payload no longer produces a dangerous aft shift. Had the pusher layout been retained, the same payload removal would have driven the CG behind the neutral point and made the aircraft unflyable without nose ballast. The configuration change forced by geometric interference therefore solved a stability problem that had not yet been discovered.

Second, the fuel tanks were positioned with their centroid at the MTOW CG. Burning the full fuel load moves the CG forward by only 0.024 $\bar{c}$, so the aircraft becomes marginally *more* stable as the mission proceeds — the safe direction for a 24-hour unmanned flight.

Third, the static margin of 31% $\bar{c}$ at MTOW is above the typical 10–20% band. This is deliberate for an unmanned platform, where a control-law failure should leave a passively recoverable airframe, but it costs trim drag and pitch responsiveness. Reducing $\bar{V}_H$ toward 0.40 would give roughly a 22% margin and save about 6 kg of tail structure; that refinement is left to the next design phase.

# 12 Requirement Compliance Analysis

The performance of the converged aircraft ($m_0 = 1464.54$ kg, $S = 27.683$ m$^2$, 220 HP installed, tractor) was evaluated against every RFP requirement, using the OpenVSP parasite-drag values and the Flow5 cruise efficiency.

## 12.1 Stall Speeds

From $V_s = \sqrt{2W_0/(\rho_0 S C_{L,max})}$ at sea level:

- **Clean** ($C_{L,max} = 1.4254$): $V_s = 24.38$ m/s = **47.4 KCAS**, against $\leq 50$ KCAS. **PASS**
- **Landing, full flap** ($C_{L,max} = 2.1274$): $V_s = 19.95$ m/s = **38.8 KCAS**, against $\leq 40$ KCAS. **PASS**

## 12.2 Climb and Ceilings

Climb performance follows from excess power, evaluated at the speed for minimum power required with the Gagg & Farrar altitude lapse applied to the turbocharged engine. All powers below are quoted as *sea-level-equivalent shaft power*, so that they may be compared directly against the 220 HP installed rating; the actual shaft power available at altitude is $\alpha$ times this figure. The residual-climb criterion used at the service ceiling is 100 fpm and at the cruise ceiling is 300 fpm.

- **Sea-level ROC:** the 1200 fpm requirement needs 182.6 HP with the refined sea-level drag (against 190.5 HP with the original assumption). With 220 HP installed, the available rate of climb is approximately **1487 fpm**. **PASS**
- **Cruise ceiling, 18,000 ft:** maintaining a 300 fpm residual climb requires 141.3 HP sea-level-equivalent. **PASS**
- **Service ceiling, 24,000 ft:** maintaining the defining 100 fpm residual climb requires 127.2 HP sea-level-equivalent. **PASS**

The service-ceiling figure is lower than the cruise-ceiling figure only because the two use different climb criteria — 100 fpm against 300 fpm. At a common criterion the required power rises monotonically with altitude.

## 12.3 Endurance and Range

Using the usable fuel of 296.99 kg from the tank quantity of 311.84 kg (the 5% trapped-fuel allowance is not counted as usable margin) and the Flow5 endurance efficiency $(L/D)_E = 27.21$:

$$E \approx 33.6 \text{ h} \tag{62}$$

against the $> 24$ h requirement — a genuine margin of 9.6 h. Flown as a transit mission the same fuel gives $R \approx 8330$ km total cruise range and an outbound/return radius of $\approx 4165$ km. **PASS**

## 12.4 Takeoff and Landing

The takeoff analysis at sea level, ISA +30°C ($\rho = 1.1095$ kg/m$^3$), with a ground roll of 200.75 m, a rotation distance of 89.19 m and an airborne segment of 86.43 m over a 50 ft obstacle:

$$S_{TO} \approx 376.4 \text{ m} \tag{63}$$

against the optional $< 2000$ m limit. **PASS**

Landing distance was evaluated with Raymer's approach method at $W/S = 10.84$ lb/ft$^2$ and $C_{L,max,L} = 2.127$:

$$S_{landing} = \frac{80(W/S)}{\sigma\, C_{L,max,L}} + 1000 = 1408 \text{ ft} = 429 \text{ m} \tag{64}$$

over a 50 ft obstacle. Landing is the more demanding of the two field cases, and both are comfortably inside the 2000 m limit.

## 12.5 Flight Envelope and Gust-Load Analysis

A formalManeuvering–gust flight envelope was constructed for the converged aircraft using the NATO AEP-4671/USAR airworthiness framework rather than retaining the previous provisional load-factor assumption [8]. AEP-4671 is the active NATO airworthiness standard for unmanned aircraft systems, and recent peer-reviewed RPA flight-envelope research identifies AEP-4671 as applicable to medium and large fixed-wing RPA in the approximately 150–20,000 kg MTOM range . The final aircraft MTOW of 1464.54 kg therefore lies inside this applicability range.

### Aircraft data used in the envelope

The final aircraft data used in the calculation are

$$m_{\mathrm{MTOW}} = 1464.54 \text{ kg}, \tag{65}$$
$$S = 27.683 \text{ m}^2, \tag{66}$$
$$b = 23.53 \text{ m}, \tag{67}$$
$$C_{L,\max} = 1.4254, \tag{68}$$
$$V_C = 70 \text{ KCAS}. \tag{69}$$

The gust mass-ratio expression requires the mean geometric chord rather than the mean aerodynamic chord. Therefore,

$$c_{\mathrm{MGC}} = \frac{S}{b} = \frac{27.683}{23.53} = 1.1765 \text{ m}. \tag{70}$$

The minimum operational mass was taken as the operating empty mass plus the required payload and the trapped fuel that remains on board:

$$m_{\mathrm{min,op}} = m_{\mathrm{OEW}} + m_{\mathrm{payload}} + m_{\mathrm{trapped}} = 1002.70 + 150.00 + 14.85 = 1167.55 \text{ kg}. \tag{71}$$

This definition is preferred to a fuel-free OEW point because the 14.85 kg trapped-fuel allowance is explicitly treated elsewhere in this report as dead weight that remains on the aircraft.

The aircraft lift-curve slope was obtained from a linear least-squares fit to the final Flow5 aircraft polar over the attached-flow interval $-4^\circ \le \alpha \le 4^\circ$:

$$\frac{dC_L}{d\alpha} = 0.10857 \text{ deg}^{-1} = 6.2207 \text{ rad}^{-1}. \tag{72}$$

The negative lift limit used to draw the negative stall boundary is

$$C_{L,\mathrm{min,prov}} = -1.2817, \tag{73}$$

obtained from the XFLR5 NACA 4415 negative-angle polar. This value is labelled *provisional*: the final Flow5 viscous 3-D analysis converged only to $\alpha = -13.75^\circ$, where $C_L = -0.93049$, before the negative stall turnover was reached. Consequently, the negative side of the envelope requires higher-fidelity 3-D validation in the next design phase.

**Maneuvering limits**

According to USAR.337, the minimum positive limit Maneuvering load factor is

$$n_{\lim,+} = \min\left(2.1 + \frac{10900}{m_{\text{MTOW}} + 4536},\, 3.8\right). \tag{74}$$

For the present aircraft,

$$2.1 + \frac{10900}{1464.54 + 4536} = 3.9165, \tag{75}$$

$$\therefore \qquad n_{\lim,+} = \boxed{3.8}. \tag{76}$$

The corresponding negativeManeuvering limit is

$$n_{\lim,-} = -0.4 n_{\lim,+} = \boxed{-1.52}. \tag{77}$$

AEP-4671 USAR.335 requires the design dive speed to satisfy

$$V_D \geq 1.25 V_C. \tag{78}$$

Using the RFP cruise speed,

$$V_D = 1.25(70) = \boxed{87.5 \text{ KCAS}}. \tag{79}$$

The AEP-4671 design speeds are defined as equivalent airspeeds. Standard compressible airspeed conversion at 20,000 ft gives

$$V_C = 69.886 \text{ KEAS}, \qquad V_D = 87.278 \text{ KEAS}. \tag{80}$$

The difference from KCAS is small at the present Mach number, but the KEAS values were used in the flight-envelope calculations.

The positive and negative aerodynamic stall boundaries are

$$n_{s,+}(V_E) = \frac{\rho_0 V_E^2 S C_{L,\max}}{2W}, \tag{81}$$

$$n_{s,-}(V_E) = \frac{\rho_0 V_E^2 S C_{L,\min,\text{prov}}}{2W}, \tag{82}$$

where $\rho_0 = 1.225$ kg/m$^3$ and $V_E$ is equivalent airspeed. At MTOW these relations give

$$V_{S,+} = 47.385 \text{ KEAS}, \qquad V_{S,-} = 49.971 \text{ KEAS}. \tag{83}$$

The theoretical positiveManeuvering corner speed is

$$V_A = V_{S,+}\sqrt{n_{\lim,+}} = 47.385\sqrt{3.8} = 92.37 \text{ KEAS}. \tag{84}$$

Since $V_A > V_D$, the MTOW positiveManeuvering boundary remains stall-limited up to $V_D$. The attainable MTOW positive load factor at $V_D$ is therefore

$$n_{+,\text{MTOW}}(V_D) = \left(\frac{87.278}{47.385}\right)^2 = \boxed{3.392}. \tag{85}$$

At the minimum operational mass, the one-g positive stall speed becomes

$$V_{S,+,\min} = 42.309 \text{ KEAS}, \tag{86}$$

and hence

$$V_{A,\min} = 42.309\sqrt{3.8} = \boxed{82.48 \text{ KEAS}} < V_D. \tag{87}$$

Thus the aircraft reaches the $+3.8g$Maneuvering limit before the dive-speed boundary in the minimum-operational-weight case.

**Discrete-gust model**

AEP-4671 uses a discrete gust criterion. Up to 20,000 ft, the prescribed derived gust velocities are

$$U_{de,C} = 15.2 \text{ m/s} \quad \text{at } V_C, \tag{88}$$

$$U_{de,D} = 7.6 \text{ m/s} \quad \text{at } V_D. \tag{89}$$

The analytical gust-load method is based on the classical Pratt formulation [9]. The aeroplane mass ratio is

$$\mu_g = \frac{2(W/S)}{\rho c_{\mathrm{MGC}} a g}, \tag{90}$$

and the gust alleviation factor is

$$K_g = \frac{0.88\mu_g}{5.3 + \mu_g}. \tag{91}$$

The resulting normal load factor is

$$n_g = 1 \pm \frac{K_g \rho_0 U_{de} V_E a}{2(W/S)}, \tag{92}$$

where $a = 6.2207$ rad$^{-1}$ and $V_E$ is in m/s.

The highest gust response in the operating altitude range occurs at approximately 20,000 ft. At that altitude,

$$\rho_{20k} = 0.65269 \text{ kg/m}^3. \tag{93}$$

For MTOW,

$$\frac{W}{S} = 518.811 \text{ N/m}^2, \tag{94}$$

$$\mu_g = 22.150, \tag{95}$$

$$K_g = 0.7101. \tag{96}$$

Equation (92) then gives

$$n^+_{g,C} = +3.850, \qquad n^-_{g,C} = -1.850, \tag{97}$$

$$n^+_{g,D} = +2.780, \qquad n^-_{g,D} = -0.780. \tag{98}$$

The raw positive $V_C$ gust point is outside the positive stall boundary and is therefore not an attainable aerodynamic state. After enforcing the stall boundary, the combined MTOW envelope has the extrema

$$\boxed{n_{\max,\mathrm{MTOW}} = +3.392}, \qquad \boxed{n_{\min,\mathrm{MTOW}} = -1.850}. \tag{99}$$

The negative critical point occurs at $V_C$.

For the minimum operational mass,

$$\frac{W}{S} = 413.602 \text{ N/m}^2, \tag{100}$$

$$\mu_g = 17.658, \tag{101}$$

$$K_g = 0.6769, \tag{102}$$

and the gust points become

$$n^{+}_{g,C} = +4.407, \qquad n^{-}_{g,C} = -2.407, \tag{103}$$

$$n^{+}_{g,D} = +3.128, \qquad n^{-}_{g,D} = -1.128. \tag{104}$$

At $V_C$, the positive raw gust point again lies outside the positive stall boundary:

$$n_{s,+}(V_C) = 2.728 < 4.407. \tag{105}$$

Thus it is clipped by aerodynamic stall. The positive combined envelope subsequently reaches theManeuvering limit at

$$V_A \simeq 82.48 \text{ KEAS}, \qquad n = +3.80. \tag{106}$$

On the negative side, the provisional negative stall boundary at $V_C$ is

$$n_{s,-}(V_C) = -2.453, \tag{107}$$

so the calculated gust point $n = -2.407$ remains just inside the provisional aerodynamic boundary and becomes the governing negative load case. Therefore, the bounding limit loads used for structural interpretation are

$$\boxed{n_{\text{limit,critical}} = +3.80/-2.41}. \tag{108}$$

Applying the conventional 1.5 limit-to-ultimate factor gives

$$n_{\text{ult},+} = 1.5(3.80) = \boxed{+5.70}, \tag{109}$$

$$n_{\text{ult},-} = 1.5(-2.407) = \boxed{-3.61}. \tag{110}$$

These values supersede the previous statement that $N_z = 3.75$ was an ultimate load factor supporting the complete envelope. The +3.8 value is a *limit* manoeuvring factor; the combined gust envelope must also be checked, particularly at reduced flight weight.

Table 20: Summary of the finalManeuvering–gust envelope.

| Parameter | MTOW | Minimum operational mass |
|---|---|---|
| Mass [kg] | 1464.54 | 1167.55 |
| $V_{S,+}$ [KEAS] | 47.385 | 42.309 |
| $V_{S,-}$ [KEAS] | 49.971 | 44.618 |
| $V_A$ [KEAS] | 92.37 ($> V_D$) | 82.48 |
| $\mu_g$ at 20,000 ft | 22.150 | 17.658 |
| $K_g$ at 20,000 ft | 0.7101 | 0.6769 |
| $n^{+}_{g,C}$ | 3.850 | 4.407 |
| $n^{-}_{g,C}$ | -1.850 | -2.407 |
| Combined $n_{\max}$ | 3.392 | 3.800 |
| Combined $n_{\min}$ | -1.850 | -2.407 |

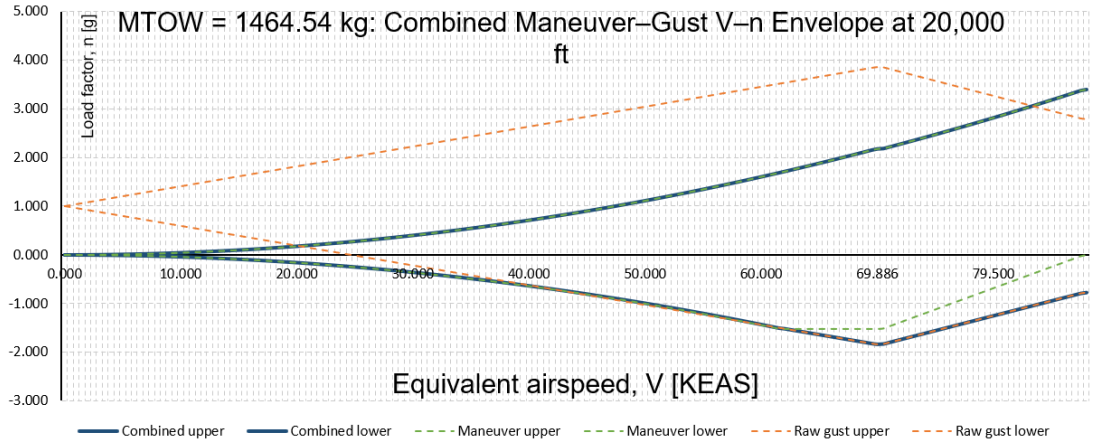



Figure 18: CombinedManeuvering–gust V–n envelope at MTOW and 20,000 ft. The raw gust lines are shown together with the stall-limited attainable envelope.

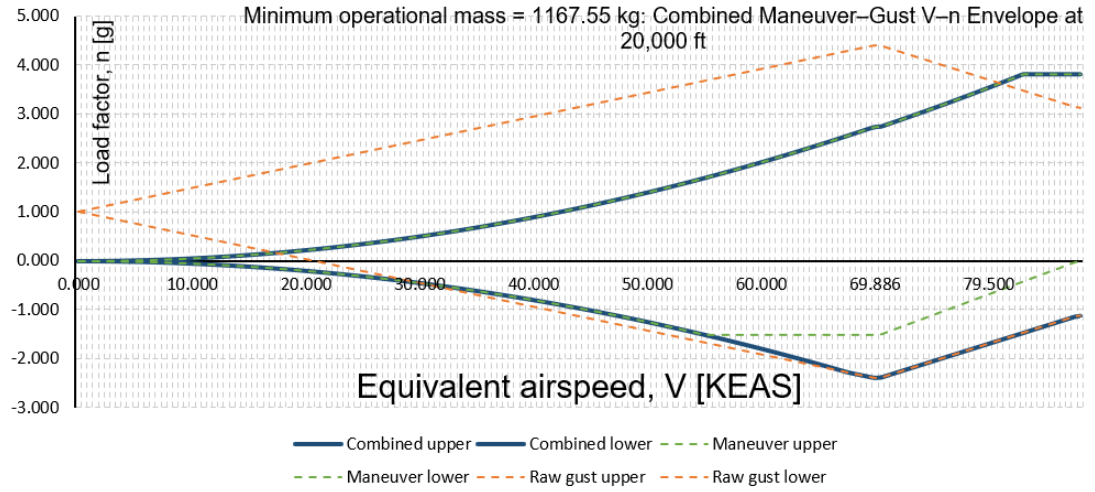



Figure 19: CombinedManeuvering–gust V–n envelope at the minimum operational mass $m = 1167.55$ kg and 20,000 ft. The negative $V_C$ gust condition is the governing negative limit-load case.

**Design implication.** The flight-envelope analysis shows thatManeuvering alone is not sufficient to define the structural load basis. At MTOW the positive side remains stall-limited below the $+3.8g$Maneuvering cap, whereas the lower-weight end-of-mission condition increases gust sensitivity and produces a governing negative load of approximately $-2.41g$ at $V_C$. The present negative boundary is intentionally conservative but remains provisional because it relies on a two-dimensional XFLR5 negative-stall limit. A higher-fidelity three-dimensional stall analysis or wind-tunnel test is therefore required before structural loads are frozen for detailed design.

## 12.6 Compliance Matrix

| Requirement | RFP Target | Computed | Margin | Status |
|---|---|---|---|---|
| Payload | $\geq 150$ kg | 150 kg | exact | **PASS** |
| Landing stall speed | $\leq 40$ KCAS | 38.8 KCAS | $-3.0\%$ | **PASS** |
| Clean stall speed | $\leq 50$ KCAS | 47.4 KCAS | $-5.2\%$ | **PASS** |
| Cruise speed | 70 KCAS | 70 KCAS analysed | — | **PASS** |
| Cruise ceiling | $> 18{,}000$ ft | 141.3 HP req. (SL-equiv.) $< 220$ HP | +56% power | **PASS** |
| Service ceiling | $> 24{,}000$ ft | 127.2 HP req. (SL-equiv.) $< 220$ HP | +73% power | **PASS** |
| Max ROC at sea level | $> 1200$ fpm | $\approx 1487$ fpm | +23.9% | **PASS** |
| Endurance | $> 24$ h | $\approx 33.6$ h (usable fuel only) | +40% | **PASS** |
| Operation radius | $> 2000$ km | $\approx 4165$ km | +108% | **PASS** |
| Total mission cruise range | 4000 km | $\approx 8330$ km | +108% | **PASS** |
| Takeoff distance (ISA +30°C) | $< 2000$ m | 376 m | $-81\%$ | **PASS** |
| Landing distance | not specified | 429 m over 50 ft | — | — |

Table 21: Final requirement compliance matrix

**Discussion of margins.** The design meets every requirement, but the margins are deliberately uneven, and this is the arithmetic consequence of constraint-driven sizing rather than an accident.

The two **stall speed** requirements are met with only 3–5% margin. That is correct: the landing stall constraint *set* the wing area at the design point, so the wing is exactly as large as physics demands to hold 1464.5 kg at 40 KCAS and no larger. A comfortable margin here would mean an over-sized, heavier wing.

The **takeoff distance** margin is enormous (81%) because the engine was sized by the sea-level climb requirement, not by takeoff. This confirms that the takeoff constraint was correctly identified as non-binding in the constraint analysis, and it is an operational bonus: the aircraft can work from short, semi-prepared strips.

The **endurance and range** margins are large for a different reason — they are the direct dividend of the drag being 36% lower than assumed. The design carries fuel for 33.6 h while needing only 24 h, but the tank has been sized to the requirement — the extra 9.6 h comes from efficiency rather than surplus fuel.

The one genuine risk item is the pair of stall speeds, which depend on the assumed single-slotted flap increment $\Delta C_{l,max} = 1.30$ and the 95% wing lift fraction. Should higher-fidelity data show lower flap effectiveness, the recovery path is to extend the flapped span from 60% to 70% of the reference area, restoring $C_{L,max}$ with no change to wing area or weight.

## 12.7 Triple Redundancy Summary

The triple-redundancy principle was applied as a cross-validation philosophy, not as a requirement to triplicate hardware. Gross weight was checked three ways: the historical regression, the component build-up, and the convergence residual. Aerodynamics was checked three ways: two-dimensional airfoil data, the OpenVSP parasite-drag build-up, and the Flow5 three-dimensional analysis — with an analytical component build-up as a fourth reference. Load factors were checked against MIL-STD-3013A, FAR-23 and competitor practice. For propulsion and flight control, redundancy is retained at subsystem

level through independent electrical, fuel/ignition and actuation paths, because adding nacelles would penalise the endurance requirement that defines the aircraft.

| **Design Item / Parameter** | **Primary Source** | **Secondary Source** | **Physical / Empirical Check** |
|---|---|---|---|
| Gross Weight Trend ($W_e/W_0$) | Raymer (UAV Model) | Sadraey (UAV Trend) | Bayraktar TB2 Operational Data |
| Engine Architecture Choice | Pugh Matrix Trade | Continental TSIO-360 Chart | Mooney M20K / Cessna 172 Benchmarks |
| Wing Taper Ratio ($\lambda = 0.45$) | Raymer Fig. 4.19 | Roskam Part II | Elliptical Lift Distribution Physics |
| Tail Volume Coefficients | Raymer Table 6.4 | Roskam Part II | Agricultural Aircraft Baseline |
| Control Surface Chord Ratios | Sadraey Ch. 12 | Raymer Chapter 6 | ICAS 2018 (Hurricane-CS Paper) |

Table 22: Triple Redundancy Cross-Validation Summary Matrix

# 13 Conclusion

This report has documented a complete conceptual design cycle for a MALE-class UAV sized against an eleven-point RFP, dominated by a 24-hour endurance requirement, a 150 kg payload, and a 40 KCAS landing stall speed. The final configuration is a 1464.5 kg, single-engine tractor aircraft with a 23.53 m span, $AR = 20$, 27.683 $m^2$ NACA 4415 wing, single-slotted flaps, a NACA 0009 T-tail, and a 220 HP turbocharged Continental TSIO-360 driving a 1.90 m three-blade propeller.

Three key engineering findings define the outcome of this work. The first is that the statistical and component-based weight estimates *agreed*: $m_{e,str}/m_0 = 0.660$ from the detailed build-up against 0.643 from the historical regression, yielding a gross weight converging from 1483.06 kg to 1464.54 kg. Two entirely independent methods landing within 1.25% of each other provides the strongest possible validation at the conceptual stage, confirming that the first-pass sizing was fundamentally sound.

The second finding was that the initial aerodynamic assumptions were overly conservative. The assumed $C_{D0} = 0.025$ proved 36% pessimistic once the actual 3D geometry was analysed in OpenVSP. This correction raised the $(L/D)_{max}$ from 22.42 to 31.42, which in turn reduced the fuel required for the 24-hour mission from 380.8 kg down to 296.99 kg (usable). That 84 kg saving is the primary reason the gross weight was able to converge slightly lower rather than forcing a significant airframe resizing. It highlights a critical conceptual design lesson: relying solely on initial assumptions without 3D geometric verification can carry large compounding errors through the sizing loop. The corollary is an honest surplus: the final aircraft design now possesses roughly 9.6 h of endurance margin over the RFP requirement, achieved entirely through aerodynamic efficiency rather than by carrying extra fuel.

The third finding involved architectural repackaging. The pusher installation, initially selected for unhindered payload field-of-view, turned out to be geometrically incompatible with the required T-tail clearance and the 1.90 m propeller, forcing a late revision to a tractor layout with a chin-mounted sensor bay. While enacted reluctantly, this change resolved a latent stability defect: moving the 238 kg engine group to the nose successfully positioned the center of gravity such that the aircraft remains stable in every loading condition, including an empty ferry flight. Had the pusher layout been retained, the payload-out CG would have shifted dangerously behind the neutral point. This configuration change, forced by a geometric clearance check, ultimately fixed a critical stability flaw—serving as a strong argument for building and checking the 3D geometry early in the design cycle.

Physics demands trades throughout the entire design process. Choosing $AR = 20$ buys the extreme aerodynamic efficiency that makes a 24-hour loiter possible, but pays for it with a heavy wing structure constituting 18% of the gross mass. Selecting a turbocharged engine secures the 24,000 ft service ceiling, but extracts a penalty in total engine group weight and thermal complexity. Placing the fuel exactly at the MTOW CG effectively eliminates CG travel during fuel burn, but restricts tank volume to the narrow confines of the inner wing box. Finally, the 22% static margin provides excellent passive stability and recoverability, but guarantees a permanent trim drag penalty. Recognizing that no design choice is entirely free is the true substance and discipline of conceptual aircraft design.

# References


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